GTM Phase 1

Phase 1 - GTM Activities for the Workato Customer Announcement

! I just want the timeline !

Key Dates

Earnings Call - mention Workato	February 24th
Stretch goal: brief media with partners	February 26th
Lighthouse announcement + Technical blog post	March 3rd

Resources

Recording
Transcript
NVIDIA Dynamo v0.4.1
Win Wire

Discussion and collaboration for the plan below in Slack:

- #internal-workato-lighthouse-gtm

What is the message we want to convey with this customer story and plan?

- The Corp Narrative articulated here
    - Specifically, we're *the* cloud for inference at scale

What are our primary objectives for this GTM?

- Highlight marquee customers and key wins, particularly technical (eg scale, performance), financial wins (eg. savings, cost predictability) and simple operations (eg. reduction in complexity, speed of getting started, etc)
- Drive AI Mindshare by showcasing our product and technical capabilities as an inferencing cloud
- Utilize our partnership with NVIDIA to amplify our message & position DO as a thought leader in AI
- Key goal is to **share PR with press ahead of launch** to maximize coverage

Secondary objective: Keep the drumbeat going on a repeatable playbook for lighthouse customer/partner activations

Who is our target audience?

- AI-native companies and digital-native enterprises running high-volume, latency-sensitive inference workloads in production (Developers, AI startups and mid-market AI builders)
- Media to build credibility with this audience
- Investors who underwrite these AI opportunities

GTM Activities: Phase 1 - Character Customer Announcement

Date / Time		STO	Status	Notes
	Prep Work			
2/6	**Interview with Workato**	Roxie Elliott	Comple…	Call getting scheduled for week of 2/2 Workato customer story assets
2/12	**Case study draft**	Roxie Elliott	In Progr…	Draft in case study tab
2/12	**Press Release: DRAFT**	Waverly Swinton	Comple…	Draft in PR tab Goal: Ship to Workato and NVIDIA no later than 2/13
2/24	**Earnings statement**	Roxie Elliott	Comple…	Workato to be referenced as one of our "eight figure deals" in the earnings call on 2/24
2/26	**PR approval Workato**	Tayler Mehit	Comple…	[EXTERNAL] Workato - Customer Story Review
2/26	**PR approval NVIDIA**	Tayler Mehit	Comple…	[EXTERNAL] NVIDIA - Workato Customer Story Revi…
2/26	**Workato name / logo release**	Roxie Elliott Ashley Armstrong Tayler Mehit	In Progr…	Completed via email Ironclad to follow
3/3	**Technical Blog Post**	Tim Kim Rithish Ramesh	Workat…	Draft done by 2/23 to share with Workato & NVIDIA teams
	Sales Enablement	Roxie Elliott	Comple…	Create slide, prep for next Rev Enablement opportunity Win Wire Template - WorkAto
	Feb 24: Earnings Announcement			
2/24	**Include reference to Workato in Earnings Call**	Melanie Strate	Comple…	Draft
	March 3: Press Announcement / Lighthouse Customer Go Live -			
3/3 9:00pm PT	**Technical Blog Post**	Person	Comple…	Workato approved 3/2 LINK: https://www.digitalocean.com/blog/workato-nvidia-technical-deep-dive-agentic-inference-cloud
3/3 6:00am PT	**Press Release**	Person	Comple…	Meghan sync with Paddy Add link Brooke to schedule release
3/3 9:00am PT	**LH customer logo updates on Web**	Waverly Swinton	Comple…	Waverly working with Rajiv to add logo to the home page

3/3 6:30-7:00a PT	**Social Media** - DigitalOcean	Jakub Lukaszc…	In Progr…	
3/3	**Social Media** - Workato	Tayler Mehit	Comple…	Social shared with Workato. Will add to results tab.
3/3	**Social Media** - NVIDIA	Tayler Mehit	Comple…	Social shared with NVIDIA. Will add to results tab.
3/3 7:00am PT	**DO ELT Social**	Jakub Lukaszc… Tayler Mehit	In Progr…	Messaging points
	Ongoing GTM			
3/11	**Case study**	Roxie Elliott	In Progr…	Publish to /customers/ page
3/24	**Video Interview** **Film 3/9**	Person	In Progr…	Film with Workato 3/9 (Workato team will film, DO team to edit)
3/4	**Joint Media Outreach**	Julie Wolf + Workato PR	Not Sta…	Outreach to start as soon as PR draft is ready *Consideration: Align outreach to pre-GTC PR (3/11)*
tbd	**GPU Product Page Update**	Waverly Swinton	Not Sta…	logo
tbd	**YouTube Video**	April Dagonese Ryan O'Connor	Not Sta…	
3/11	**Technical Blog Post** (on DO Site)	Tim Kim Rithish Ramesh	Not Sta…	Publish to blog LC GTM Plan: Workato - Q2 2026
3/11	**Technical Blog Post** *(on Workato site)*	Rithish Ramesh Roxie Elliott	Not Sta…	
3/11	**Technical Social Post**	Jakub Lukaszc…	Not Sta…	Aligned with above blog LC GTM Plan: Workato - Q2 2026
tbd	**AI Newsletter**	Waverly Swinton	Not Sta…	✅Newsletter send date now aligned with announcement (Wed) Waverly/Roxie working on copy
tbd	**Workato GTC Party**	@person	Not Sta…	Sending a team member to Workato's GTC party. To amplify on Social

Results and Retrospective

Results and Retrospective

Summary of Objectives from the GTM Plan

What was the message we wanted to convey with this customer story and plan?

- Workato AI Lab's drive increased success for their customers due to optimized NVIDIA Dynamo + H200 GPUs all on DigitalOcean's Inference cloud. Deep collaboration and support from DigitalOcean teams made these results possible.

What were the primary objectives for this PR

- Our objective is to share the customer story widely for press and investor pickup
-

Did we have any secondary objectives?

Results & Impact

Item	Owner	Output	Metrics	Summary / Impact Analysis
Press Release	Brooke Meghan	Business Wire Workato Press Cover…	According to BusinessWire, there were a total of 74 syndicated postings, reaching a potential audience of at least 60.5M. We've included the top syndications, such as Yahoo Finance and Associated Press in our coverage tracking. BusinessWire did not catch AP, so that would add another 41M.	
Social Media - DigitalOcean	Jakub Julia	LinkedIn X Threads Instagram Bluesky	•	
Social Media - NVIDIA	Tayler			

Social Media - Workato	Tayler Roxie	Workato shared on X (3/3)“We are at SKO - so our social play will be a follow-up over the week/next week.”		
Website Update	Evan			
AI Newsletter	Waverly			

Retro & Learnings

What worked? What went well?

What could be improved?

What were key learnings and takeaways?

Actions

- [ ]

Earnings Statement

STO: Roxie Elliott
Contributors: Waverly Swinton Tayler Mehit
Go Live Date / Time: Feb 24, 2026
Reminder: 2/16 is a US holiday

Reviewer / Activity	Approved?	Send	Deadline	Notes/Comments
Meghan Grady	Approved	2/12	2/13	
Melanie Strate	In progress	2/12	2/13	
Laura Schaffer	Not started			
Workato review	Approved	2/13	2/19	[EXTERNAL] Workato - Customer Story R…
NVIDIA approval (N/A) no NVIDIA mention	Not started	2/13	2/19	[EXTERNAL] NVIDIA - Workato Customer …

Our momentum in the AI space continues to accelerate, highlighted by a new seven-figure deal with Workato, a leader in enterprise agentic orchestration, who chose DigitalOcean's NVIDIA H200 GPUs and managed Kubernetes (DOKS) to power their AI Lab. Workato cited our speed of deployment and hands-on technical support as the primary drivers for selecting DigitalOcean over hyperscalers. They have achieved a 2x improvement in time-to-value and 67% lower cost per 1 million tokens for training their proprietary agentic models. Our collaboration with Workato, demonstrates how DigitalOcean's specialized, high-performance infrastructure delivers the speed, performance, and cost predictability that Digital Native Enterprises need to compete in AI research.

Quote: "DigitalOcean stood out to us through sheer speed and commitment. That level of trust and responsiveness was a deciding factor. The support team has been exceptional. With the hyperscalers, we'd probably still be waiting in a queue."

Win wire

FINAL - Draft PR

COPY LOCKED - Mar 2

STO: Meghan Grady
Contributors: Waverly Swinton Tayler Mehit Julie Wolf Meghan Grady
Go Live Date / Time: **March 3 at 6:00am PT**
Copy lock:
Asks for Workato: Confirm timescale for millions of workflows Rithish Ramesh

LIVE LINK - BIZ WIRE

Reviewer / Activity	Approved?	Send	Deadline	Notes/Comments
Meghan Grady	Approved	2/12	2/13	Approved - 2/22
Julie Wolf	Approved	2/12	2/13	
Nihar Namjoshi	Approved	2/13	2/13	
Karnik Modi	Approved	2/17	2/19	
Laura Schaffer	Approved	2/17	2/19	
Legal - Tiffany Hui	Approved	2/19	2/26	
Wade Wegner	Approved	2/24	2/25	
Melanie Strate	Approved	2/24	2/25	
Paddy Srinivasan	Approved	2/24	2/26	
Workato review	Approved	2/13	2/26	Workato has finished its Marketing Review. Now in Technical, and Exec review.
NVIDIA approval	Approved	2/13	2/26	Approval received, edits incorporated
Final copy edits	Not started	2/26	3/2	File
Story released to Business Wire	Not started	N/A	Monday, 3/2 4pm ET	

DigitalOcean Powers Workato's Agentic Enterprise with Production-scale AI

Workato's AI Research Lab achieves 67% higher throughput, 77% faster time-to-first-token, and 67% lower inference costs on DigitalOcean's Agentic Inference Cloud, powered by NVIDIA

Broomfield, Colo.--(BUSINESS WIRE)--Mar 3, 2026 – DigitalOcean (NYSE: DOCN), the Agentic Inference Cloud built for production AI, today announced that Workato's AI Research Lab is using its vertically integrated, inference-optimized platform, accelerated by NVIDIA Hopper GPUs, to advance the development of its next-generation enterprise AI agents while materially improving performance, cost efficiency, and time to value.

After moving its AI Labs workloads to DigitalOcean, Workato achieved immediate gains for frontier models, including Llama-3.3-70B:

- **Inference cost:** $0.77 power 1M tokens – **67% lower**
- **Throughput:** 13,561 tokens per second per GPU – **67% higher**
- **Time-to-First-Token** (TTFT): 1,455 ms at high load – **77% faster**
- **Time-to-Value:** Reduced from weeks to days – **2X+ acceleration**

With thousands of customers globally deploying over 1 trillion tasks, the Workato ONE platform enables customers to build, deploy, and govern AI agents at an enterprise scale. Built on a decade of integration expertise spanning 14,000+ applications, Workato's platform enables organizations to move from simple automation to agentic AI that can reason, act, and orchestrate work across the entire business.

To support that vision, Workato AI Research Lab required infrastructure capable of handling distributed training and sustained, reasoning-heavy inference under real production load. Not only was DigitalOcean able to provide high-performance NVIDIA Hopper GPUs faster than any other provider for the team to begin their work, but the Workato AI Research Lab team quickly discovered the significant performance boost and improved total cost of ownership as a result of DigitalOcean's inference-optimized architecture and simplified experience.

"Before DigitalOcean, we didn't have a dedicated solution for in-house training and multi-node serving, and that was a major blocker for AI research," said Oscar Wu, AI Research Scientist at Workato. "DigitalOcean was the fastest provider to get us up and running, enabling us to advance our AI programs. The collaboration on performance optimization coupled with the support from the DigitalOcean team of solutions architects, accelerated our progress by roughly two to three times."

Working closely with Workato, DigitalOcean helped design and tune a distributed inference architecture on DigitalOcean Kubernetes (DOKS). As part of this collaboration, DigitalOcean configured NVIDIA Dynamo to intelligently coordinate workloads across interconnected GPU clusters. This ensured requests were routed to the most efficient compute resources in real time, reducing redundant processing, lowering costs, and improving responsiveness under heavy demand.

By optimizing orchestration around the model and intelligently routing requests across interconnected NVIDIA Hopper clusters, the platform eliminated redundant computation, a primary cost driver for long-context AI workloads. The result was sustained throughput, significantly faster time-to-first-token, and materially improved price-performance under high concurrency.

"As AI adoption accelerates, inference at scale is becoming the defining challenge for the industry," said Dave Salvator, Director of Accelerated Computing Solutions at NVIDIA. "The integration of the NVIDIA accelerated computing platform with DigitalOcean's inference-optimized platform unlocks the full potential of production-scale AI. The significant performance gains achieved by Workato highlight the impact of this collaboration."

For AI companies scaling production, inference economics directly impact margins, making cost efficiency and predictability critical. By moving to DigitalOcean's optimized environment, Workato reduced inference costs by 67% to $0.77 per million tokens while achieving a 33% hardware price-performance advantage.

Equally important, DigitalOcean's managed Kubernetes environment abstracted control-plane complexity and GPU scheduling, allowing Workato's lean AI Labs team to focus on research and product development rather than infrastructure management. This vertically integrated approach, spanning hardware, orchestration, and networking, is critical to reduce operational overhead while improving performance consistency and cost predictability.

"DigitalOcean lets us focus on research and advancing our models instead of managing infrastructure," said Kevin Huang, Infrastructure Engineer at Workato AI Labs. "We can provision GPUs quickly, deploy inference workloads in production, and iterate on real customer traffic without getting bogged down in platform complexity. The speed and performance has been critical to maintaining our momentum."

"Workato is pushing the frontier of agentic enterprise software," said Paddy Srinivasan, CEO of DigitalOcean. "As AI companies move from experimentation into production, the winners will be those who can iterate quickly on real customer workloads. We're proud to support Workato's momentum by providing an inference-optimized environment that lets their team focus on shipping — not managing infrastructure."

DigitalOcean is building the Agentic Inference Cloud for production AI, partnering with ambitious AI-native companies and enabling them to operate inference reliably at scale with predictable economics. Learn more about the DigitalOcean Agentic Inference Cloud.

About DigitalOcean

DigitalOcean is the Agentic Inference Cloud built for AI-native and Digital-native enterprises scaling production workloads. The platform combines production-ready GPU infrastructure with a full-stack cloud to deliver operational simplicity and predictable economics at scale. By integrating inference capabilities with core cloud services, DigitalOcean's Agentic Inference Cloud enables customers to expand as they grow — driving durable, compounding usage over time. More than 640,000 customers trust DigitalOcean to power their cloud and AI infrastructure. To learn more, visit www.digitalocean.com

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Deprecated [v2] Draft PR

STO: Waverly Swinton
Contributors: Waverly Swinton Tayler Mehit Julie Wolf TBD
Go Live Date / Time: **March 3 at [TIME]**
Copy lock:
Asks for Workato: Confirm timescale for millions of workflows Rithish Ramesh

OUT OF DATE - use V3

DigitalOcean Powers Workato's Agentic Enterprise with Production-scale AI

Workato achieves 50% higher throughput and 2–3X faster deployment on DigitalOcean's Agentic Inference Cloud, accelerated by NVIDIA GPUs

Broomfield, Colo.--(BUSINESS WIRE)--Mar xx, 2026 – DigitalOcean (NYSE: DOCN), the Agentic Inference Cloud built for production AI, today announced that Workato is using its inference-optimized platform, accelerated by NVIDIA HGX™ H200 GPUs to advance the development of its next-generation enterprise AI agents.
After moving its AI Labs workloads to DigitalOcean, Workato increased AI throughput by 50%, reaching 14,294 tokens per second per GPU, and reduced time to deploy new models by 2–3X. The company also lowered training costs per million tokens by 67%, while enabling reliable multi-node training and sustained inference for mission-critical enterprise workloads.

Workato, which serves tens of thousands of customers globally and deploys millions of automated workflows [a week], recently launched Workato One, a unified platform for building, deploying, and governing AI agents at enterprise scale. Through enterprise Model Context Protocols (MCPs), AI Copilot, and autonomous workflow agents, Workato is advancing what it calls the "Agentic Enterprise," where AI systems execute work across IT, HR, finance, and sales.

Before partnering with DigitalOcean, AI Labs' efforts were limited by lack of GPU capacity. Workato evaluated multiple providers but encountered long wait times and complex quota processes that delayed research progress. DigitalOcean provided high-performance GPUs faster than any other provider the team evaluated, letting Workato begin training and inference workloads immediately without lengthy provisioning delays.

Workato's AI Labs needed infrastructure for inference optimization but were blocked by long wait times and complex procurement from multiple GPU providers. DigitalOcean quickly provided a hands-on support team that not only accelerated their go-to-market but also delivered GPUs optimized for real-time AI workloads.

"Before DigitalOcean, we didn't have a dedicated solution for in-house training and multi-node serving, and that was a major blocker for AI research," said Oscar Wu, AI Research Scientist at Workato. "DigitalOcean was the fastest provider to get us up and running with H200 capacity, moving us from constrained to fully capable of advancing our AI programs. The speed of access, coupled with the support from their engineering team, accelerated our progress by roughly two to three times."

Workato runs interconnected NVIDIA HGX™ H200 clusters managed by DigitalOcean Kubernetes (DOKS), enabling multi-node model training using frameworks such as Megatron Bridge and customized NVIDIA software packages. The infrastructure supports proprietary distributed inference and training systems built by Workato AI Labs.

“DigitalOcean is the only provider that lets me focus on research instead of managing infrastructure,” said Kevin Huang, Infrastructure Engineer at Workato AI Labs. “We can provision GPUs quickly, deploy inference workloads in production, and iterate on real customer traffic without getting bogged down in platform complexity. That speed has been critical to maintaining our momentum.”

As AI applications move from experimentation to production, inference becomes the operational backbone of value creation. Agentic systems generate sustained token workloads, require low latency under load, and must maintain cost predictability as usage scales. DigitalOcean’s Agentic Inference Cloud is purpose-built for this production phase of AI, combining inference-optimized GPU compute, managed Kubernetes orchestration, integrated networking, and predictable usage-based economics. The platform delivers high performance at scale with radically simple operations, enabling teams to focus on building AI products rather than managing infrastructure complexity.

Workato is pushing the frontier of agentic enterprise software,” said Paddy Srinivasan, CEO of DigitalOcean. “As AI companies move from experimentation into production, the winners will be those who can iterate quickly on real customer workloads. We’re proud to support Workato’s momentum by providing an inference-ready environment that lets their team focus on shipping — not managing infrastructure.”

NVIDIA HGX™ H200 GPUs power the interconnected clusters running on DigitalOcean, providing the compute foundation for Workato’s distributed training and inference workloads. “Reasoning-heavy agent systems require robust, high-performance infrastructure,” said David Salvator, Director of Accelerated Computing Solutions at NVIDIA. “By leveraging NVIDIA HGX™ H200 GPUs within DigitalOcean’s Kubernetes environment, Workato is accelerating enterprise AI innovation.”

DigitalOcean is building the cloud for production AI — partnering with ambitious AI-native companies and enabling them to operate inference reliably at scale with predictable economics. As agentic AI becomes embedded across enterprise software, production-grade inference will define the next phase of cloud computing.

Learn more about the DigitalOcean Agentic Inference Cloud.

About DigitalOcean

DigitalOcean is the Agentic Inference Cloud built for AI-native and Digital-native enterprises scaling production workloads. The platform combines production-ready GPU infrastructure with a full-stack cloud to deliver operational simplicity and predictable economics at scale. By integrating inference capabilities with core cloud services, DigitalOcean’s Agentic Inference Cloud enables customers to expand as they grow — driving durable, compounding usage over time. More than 640,000 customers trust DigitalOcean to power their cloud and AI infrastructure. To learn more, visit www.digitalocean.com

Deprecated [v1] Press Release Draft

STO: Waverly Swinton
Contributors: Waverly Swinton Tayler Mehit Julie Wolf TBD
Go Live Date / Time: **March 3 at [TIME]**
Reminder: 2/16 is a US holiday
Copy lock:

OUT OF DATE - use V3

DigitalOcean's Agentic Inference Cloud, Accelerated by NVIDIA GPUs, Increases Throughput and Deployment Speed for Workato's Agentic Workloads

Workato achieved a 50% increase in production throughput and 2x deployment speed with DigitalOcean's optimized inference architecture

Broomfield, Colo.--(BUSINESS WIRE)--March 3, 2026 – DigitalOcean (NYSE: DOCN), the Agentic Inference Cloud built for production AI at scale**,** today announced that through optimized architecture and NVIDIA HGX™ H200 GPUs, its Agentic Inference Platform has achieved a 50% throughput increase–surpassing 14,000 tokens per second–for enterprise automation leader Workato.

Workato has selected GPU Droplets accelerated by NVIDIA HGX™ H200 and DigitalOcean Managed Kubernetes (DOKS) to power its specialized AI Lab, moving away from hyperscalers to gain superior speed, predictable cost, and technical support. Following DigitalOcean's recent high-profile collaboration with Character.AI, this achievement underscores the company's growing momentum as a strategic partner for managing complex inference workloads at scale.

Accelerating AI Breakthroughs: How Workato Improved Inference Performance with DigitalOcean's Specialized Infrastructure

Workato helps over 12,000 global customers unify data, processes, and applications. In 2025, the company launched Workato One, a unified platform for the "Agentic Enterprise," where autonomous AI agents execute complex business functions across IT, HR, finance, and sales. Workato AI Labs, based in San Francisco, serves as the company's primary engine for AI breakthroughs, focusing on the development of autonomous agents via synthetic evaluation frameworks, automated reinforcement learning, and custom model optimization. To power these breakthroughs, Workato required an infrastructure partner that could move at the speed of research.

After evaluating several other providers, including leading hyperscalers and AI neoclouds, Workato chose DigitalOcean's Agentic Inference Cloud for its superior speed of execution and the specialized expertise of its engineering team. DigitalOcean helped Workato supercharge the performance of next-generation Llama models by combining specialized hardware with innovative software optimizations to deliver faster, more efficient AI processing without losing accuracy. This partnership allowed Workato to build a

dedicated infrastructure that could handle both intensive model training and high-throughput inference with a level of agility the hyperscalers couldn't match.

By migrating its inference and fine-tuning workloads to DigitalOcean's high-performance, dedicated infrastructure, Workato achieved:

- **50% throughput increase**, reaching 14,294 tokens/sec per GPU from the baseline of 9,540 tokens/sec.
- **67% lower cost** per one million tokens for training proprietary agentic models.
- **Reliable low latency**, delivering near-instant interactions even when running massive, complex AI models.

"DigitalOcean stood out to us through sheer speed and commitment," said **Oscar Wu, AI Research Scientist and Founder of AI Labs**. "That level of trust and responsiveness was a deciding factor. The support team has been exceptional, providing professional assistance for complex AI workloads. With the hyperscalers, we'd probably still be waiting in a queue."

From Constraints to Scale: Optimizing Inference Workloads with NVIDIA GPUs and DigitalOcean Kubernetes

As a result of these performance gains, Workato has transitioned from being resource-constrained to fully capable of advancing in-house research. Now, Workato utilizes DigitalOcean's compute and Kubernetes orchestration layers to run customized NVIDIA software packages, building proprietary distributed inference and training frameworks.

"DigitalOcean is the only provider that lets me focus on my job instead of fighting infrastructure," added **Kevin Huang, AI Lab Engineer at Workato**. "The user interface is incredibly intuitive, allowing us to manage complex environments autonomously, while their expertise in securing cutting-edge hardware—like the B300 nodes for our upcoming NVFv4 testing—saves us a massive amount of time."

"The era of agentic AI requires a fundamental shift in how we build and deploy infrastructure, as reasoning-heavy workloads demand significantly more computational power than traditional chatbots," said **David Salvator, Director of Accelerated Computing Solutions at NVIDIA**. "By leveraging the NVIDIA HGX™ H200 platform and DigitalOcean's integrated Kubernetes environment, Workato is at the forefront of this transition—transforming complex inference into a competitive advantage through superior speed and architectural efficiency."

Sustained Momentum for DigitalOcean's Agentic Inference Cloud

Workato's results are the latest evidence of DigitalOcean's market differentiation-its ability to optimize both hardware and software for customers. By offering the latest NVIDIA hardware paired with hands-on technical partnership and a full stack offering, DigitalOcean is positioning itself as the essential alternative to the complexity of hyperscalers and the technical limitations of AI neoclouds.

"Digital native enterprises don't just need GPUs; they need a partner that eliminates infrastructure friction. We are proud to help Workato lower their costs and accelerate their path to production" said **Paddy Srinivasan, CEO of DigitalOcean. "**Workato is a pioneer in the agentic orchestration space, and their

success demonstrates how DigitalOcean is providing high-performance at scale with predictable cost efficiency."

By removing the traditional barriers to entry—prohibitive costs and infrastructure complexity—DigitalOcean ensures that high-growth, AI-native enterprises like Workato can focus on building intelligent agents rather than managing infrastructure. As the demand for agentic AI continues to scale, DigitalOcean remains committed to providing the performance of a hyperscaler with the simple operations customers have come to expect.

About DigitalOcean

DigitalOcean is the Agentic Inference Cloud built for AI-native and Digital-native enterprises scaling production workloads. The platform combines production-ready GPU infrastructure with a full-stack cloud to deliver operational simplicity and predictable economics at scale. By integrating inference capabilities with core cloud services, DigitalOcean's Agentic Inference Cloud enables customers to expand as they grow — driving durable, compounding usage over time. More than 640,000 customers trust DigitalOcean to power their cloud and AI infrastructure. To learn more, visit www.digitalocean.com

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FINAL - Technical Blog

STO: Roxie Elliott
Contributors: Roxie Elliott Waverly Swinton Tayler Mehit Rithish Ramesh Tim Kim Karnik Modi
Go Live Date / Time: **March 11 at [TIME]**
URL: https://www.digitalocean.com/blog/workato-nvidia-technical-deep-dive-agentic-inference-cloud
Reminder: 2/16 is a US holiday

Reviewer / Activity	Approved?	Send	Deadline	Notes/Comments
Karnik Modi	Approved		2/19	
Nihar Namjoshi	Approved		2/19	Dates will waterfall from first draft
Tim Kim Roxie Elliott	Approved			
Piyush Srivastava	Approved			
Meghan Grady	Approved	2/19	2/23	
Laura Schaffer	Approved	2/19	2/23	
Legal - Tiffany	Approved	2/19	2/26	
Workato review	Approved	2/23	3/5	[EXTERNAL] Workato - Customer Story R…
NVIDIA approval	Approved	2/23	3/5	Approved by Elijah + Dynamo team 2/27 [EXTERNAL] NVIDIA - Workato Customer …

Authors: Rithish Ramesh, Piyush Srivastava, Karnik Modi, Tim Kim

How DigitalOcean's Agentic Inference Cloud powered by NVIDIA GPUs Achieved 67% Lower Inference Costs for Workato

Workato's AI Research Lab is focused on helping customers extend their production automation with agentic AI capabilities, systems that can reason, act, and orchestrate work across the business. At Workato's scale, processing 1 trillion automated workloads, LLM inference efficiency is a hard requirement: every millisecond of latency and every wasted GPU cycle directly impacts cost, throughput, and reliability. To make agentic workloads production-ready, the team needed an inference stack built for production scale – delivering predictable performance and unit economics at scale, not just raw compute.

DigitalOcean partnered with Workato's AI Research Lab team to design and tune this deployment on its Agentic Inference Cloud, using NVIDIA Dynamo with vLLM on DigitalOcean

Kubernetes Service (DOKS). To support 100K-token context lengths without degrading performance, NVIDIA H200 GPUs were selected for their 141GB HBM3e memory capacity.

The memory footprint of the workload was around 125 GB (comprising the model weights, key value cache, and activation buffer), so a single NVIDIA H200 GPU is able to fit the whole footprint. However, the team used 8-way tensor parallelism per node to maximize sustained throughput and latency stability under a concurrent load.

Digitalocean tested across two different configurations for Workato, and afterwards, the results for NVIDIA Dynamo + vLLM on DOKS showed:

- **Best in class queries-per-second** across all tested configurations
- **67% higher throughput per GPU** with **79% lower end-to-end latency** and **77% time-to-first-token** compared to different configurations on identical hardware
- **33% lower hardware cost** using a NVIDIA H200 GPU vs. a NVIDIA A100 GPU for equivalent performance
- **67% lower model cost** while using half the GPUs

The key here was to introduce key/value (KV)-aware routing in order to reduce redundancies and capture maximum value across performance and cost for the inference stack.

How LLMs Process Requests and Why It Gets Expensive at Scale

Before getting into the architecture decisions, it's worth understanding the mechanics that drive inference cost and why this is a complex problem that Workato needed to solve. Every LLM inference request goes through two phases:

- **Prefill** is where the model processes the entire input prompt and builds up internal memory, called key/value (KV) states, for every token it has read. This phase is compute-heavy and scales quadratically ($O(n^2)$) with input sequence length. For long-context workloads (e.g 10K-100K token prompts), prefill can consume the majority of total inference cost.

  The primary reason for this is that the model needs to compute self-attention scores for every token against every other token in the prompt. As an example, if the prompt is 1000 tokens, the model performs roughly 1000 * 1000 attention operations. If the prompt is 100,000 tokens (as the case with Workato's workload), those operations jump to 10 billion. 100K token prefills require many floating point operations per second (FLOPs) and it can take several seconds of 100% GPU utilization, resulting in lower throughput per GPU directly contributing to cost.
- **Decode** is where the model generates tokens one at a time, using those cached KV states to predict each next token. This phase is memory bandwidth bound; performance of the decode phase directly impacts token streaming latency.

There are real-world workloads that share common input prefixes where a significant, identical “block” of text is reused across multiple requests. In enterprise SaaS applications (like Workato’s AI Research Lab), there is often a high degree of prefix sharing across inference requests. As the GPU does prefill operations, it builds in-memory context (KV cache) which is expensive to build specifically for long-prompt workloads.

Now, if subsequent queries are all routed to separate GPUs, every GPU has to re-build the KV cache, resulting in redundant FLOPs being consumed which could have instead been used to serve other queries.

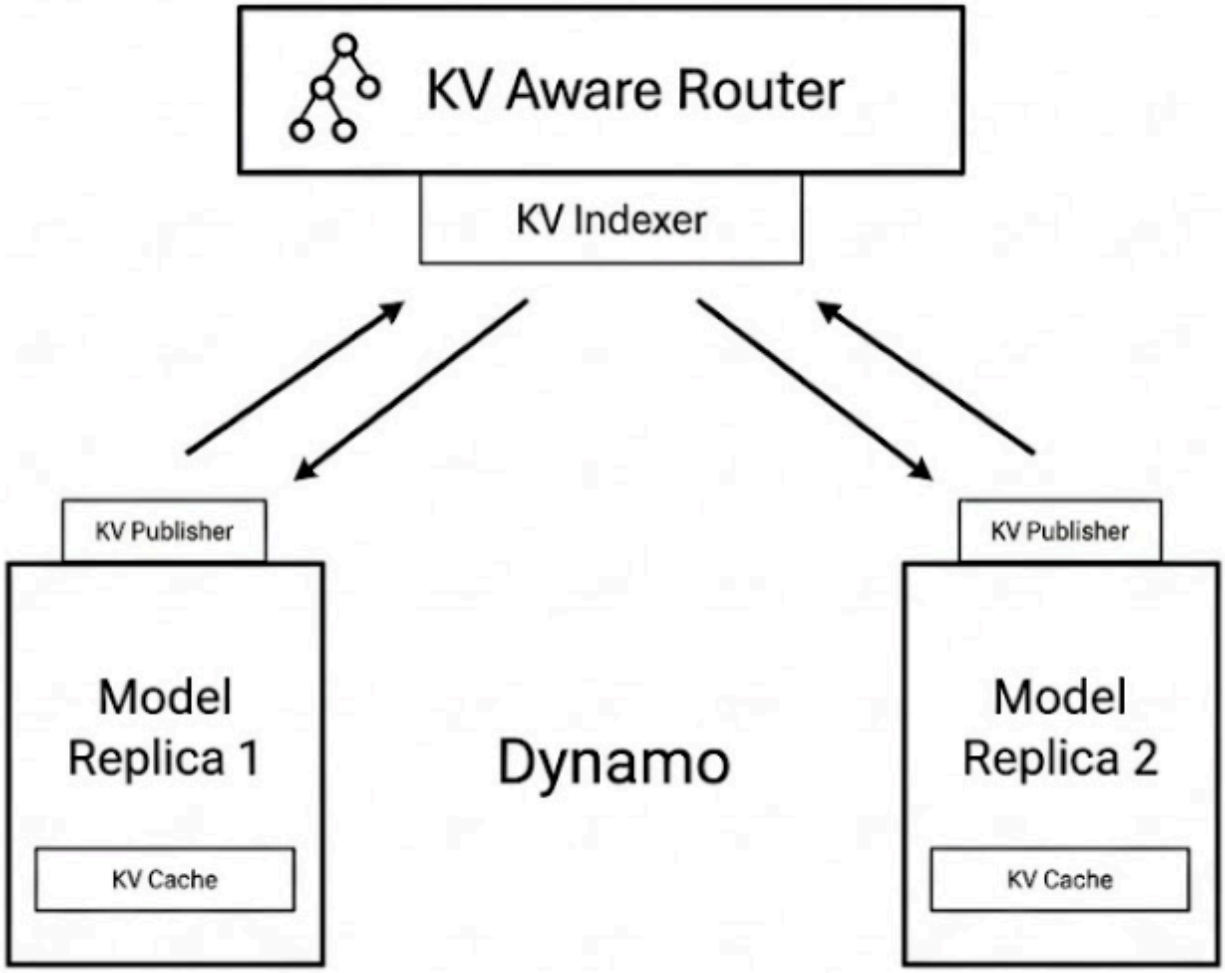
How KV-Aware Routing Addresses the Problem

KV-Aware routing is a technique which utilizes the commonality of prefixes and routes them to the same GPU. This helps by enabling the GPU to leverage a warm KV cache (often via RadixCache) to skip the compute-heavy prefill phase entirely.

This helps in dramatically reducing first token latency (TTFT) for the end user, and significantly increases the total throughput of the cluster by reclaiming GPU FLOPs which would have otherwise been spent on redundant prefill computations.

NVIDIA Dynamo with DOKS: The Orchestration Brain for KV-Aware Routing

NVIDIA Dynamo is an open-source, low-latency, modular inference framework designed to operate on top of individual inference engines. It is engine-agnostic and can orchestrate backends like vLLM (this is what we used here), TensorRT-LLM, and SGLang. Dynamo is not designed to make a single GPU faster. It is designed to prevent the cluster from doing redundant work and keep the right GPUs busy with the right phase of inference. In the context of Workato, we used Dynamo for its KV-aware routing capabilities.



NVIDIA Dynamo transforms standard LLM infrastructure by introducing a sophisticated orchestration layer that far exceeds the capabilities of a vanilla multi-node setup. At its core, Dynamo functions as a global scheduler with a comprehensive view of every GPU in the cluster, moving beyond the limitations of workers that only see their own local resources. This global perspective is managed by a cluster-level KV cache manager that meticulously tracks which tokens reside on specific workers, identifies which blocks are hot or evictable, and determines the optimal time to reuse, offload, or recompute various cache segments.

The defining feature of this architecture is the KV-aware router, which replaces traditional, "blind" round-robin distribution with LLM-aware request routing. Rather than treating all workers as equal, the router utilizes a complex cost function to score candidate workers based on existing cache overlaps and the critical trade-offs between system metrics. Specifically, it balances the need to minimize prefill costs-improving Time to First Token (TTFT)-against the requirements for efficient decode performance, such as Inter-Token Latency (ITL) and Time Per Output Token (TPOT). By integrating this real-time awareness of global cache states, Dynamo ensures that routing decisions are globally optimal, enabling seamless KV cache offloading and maximum reuse across disparate requests.

However, deploying a cluster-aware inference stack like NVIDIA Dynamo introduces real operational complexity. You're no longer running a few vLLM servers behind a load balancer. You're running a distributed system with a routing frontend, instrumented worker backends, a KV cache manager, and real-time coordination between all of them. Getting that right requires both the right framework and platform powering it all. While NVIDIA Dynamo provides the logic, DOKS provides the execution environment that makes KV-aware routing possible. DOKS doesn't just 'run' the pods; its native service discovery and intelligent scheduling allow the Dynamo frontend to act as a cluster-wide dispatcher. This ensures that incoming requests are instantly routed to the specific GPU node where the relevant KV cache already lives,

This eliminates redundant computations and capture maximum value from the full inference stack, specifically:

- **Independent replica management** for the routing frontend and worker backends means the team could scale each layer separately as they tuned configurations.
- **Kubernetes-native service discovery** lets the router frontend find and route to worker pods cleanly, without custom networking work.
- **Operational simplicity**: DOKS abstracts away the control plane entirely, so Workato's engineers can focus on inference optimization, instead of Kubernetes orchestration.

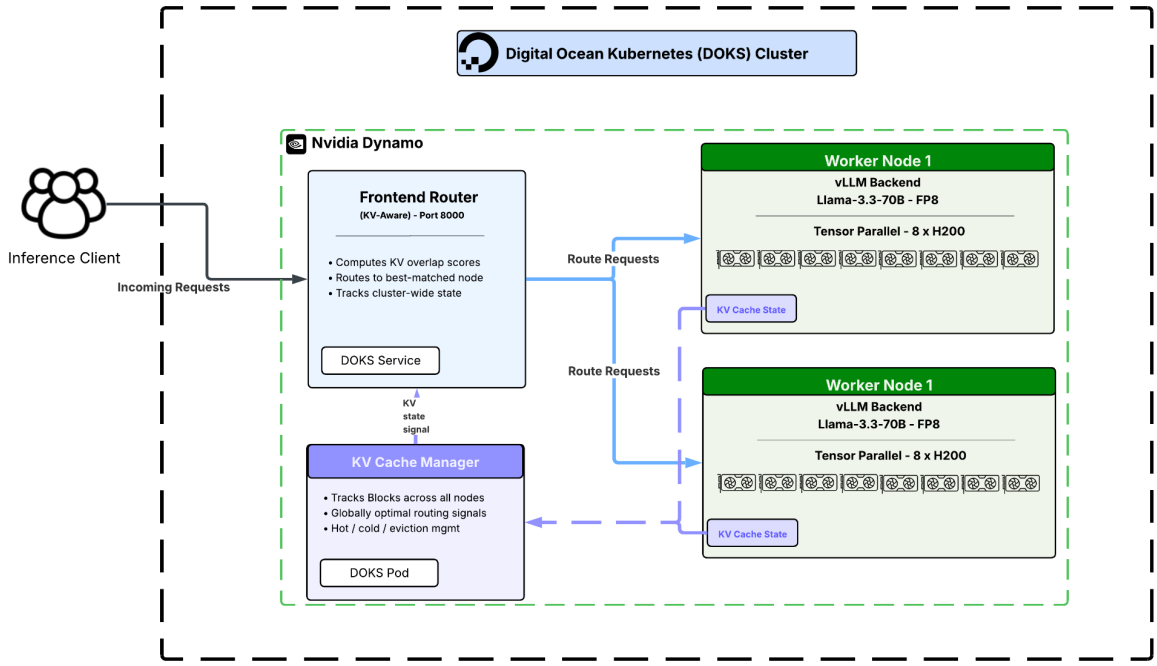
Inference Stack Architecture

We ran the evaluation across a variety of DOKS clusters using **nvidia/Llama-3.3-70B-Instruct-FP8** as our model**:**

- 1-node with 8 NVIDIA H200 GPUs
- 2-node with 16 GPUs
- 4-node with 32 GUs

We also compared each topology for NVIDIA Dynamo + vLLM (chosen due to its broad feature coverage and clean integration surface) across two configurations: one with no KV-aware routing and one with KV-aware routing enabled. This would give Workato a real map of the performance-to-cost trade off space, not just a single data point.

The full cluster architecture for the 2-node deployment looked like this:



The frontend router of NVIDIA Dynamo is a lightweight DOKS service sitting in front of the GPU worker pods. The workers are full vLLM instances instrumented with Dynamo agents that continuously publish KV cache state back to the router and the cache manager. The KV cache manager is a separate pod that maintains the global view, tracking what's hot on which node, signaling evictions, and feeding the router's scoring function with fresh state. Separating the router, workers, and cache manager as distinct DOKS workloads allowed Workato to tune each layer independently and enabled the architecture to scale horizontally as workload demands grow.

The Two Configurations Tested

Configuration 1: No KV-Aware Routing

This configuration used the same model, same GPU, but with vLLM's own optimizations enabled: prefix caching, chunked prefill, FP8 weights and KV cache. This **does not** include KV-aware cluster routing—each node makes smart local decisions, but there is no global view of cache state across the cluster.

```shell
vllm serve nvidia/Llama-3.3-70B-Instruct-FP8 \
  --tensor-parallel-size 8 \
  --enable-prefix-caching \
  --enable-chunked-prefill \
  --quantization fp8 \
  --kv-cache-dtype fp8 \
  --trust-remote-code
```

What actually happens when this configuration was put under a 100K-token prefill load? When two prompts share a long common prefix but land on different workers, both nodes redundantly recompute the same prefill because neither is aware of the work the other has already performed. Under this scenario, cache locality is entirely accidental, meaning that whether a request benefits from a warm KV cache is a matter of pure luck rather than strategic placement. This lack of coordination causes severe load imbalances to emerge; some workers become bogged down by long prefills while others manage decode-heavy traffic, causing their processing queues to diverge. Ultimately, tail latency explodes at higher concurrency levels because a single unlucky routing decision can pin a specific worker with multiple heavy prefills while other available resources sit idle.

Configuration 2: KV-Aware Routing

This is where the architecture fundamentally changes and out of the two configurations we tested, this was the clear winner. At a high level, the NVIDIA Dynamo deployment utilizes cluster-aware routing and splits the system into 1) Frontend (router + scheduler), 2) Set of workers (vLLM instances instrumented with Dynamo)

Dynamo Frontend (KV Router Enabled)

```none
python3 -m dynamo.frontend --http-port 8000 --router-mode kv
```

The NVIDIA Dynamo frontend actively computes KV overlap scores against available workers and applies a specialized routing cost function to decide precisely where prefill and decode phases should execute. By tracking cluster-wide load and the current global cache state, the system ensures that every request is handled with maximum efficiency. In this setup, the frontend is being run with KV routing fully enabled to validate these architectural improvements.

Workers (deployed as DOKS pods on GPU nodes):

Each worker is a vLLM instance launched via NVIDIA Dynamo's wrapper. The integration exposes -

- **KV cache events:** What blocks are created, reused, or evicted
- **Metrics:** Queue depth, decode load, memory pressure

Recipe: Dynamo + vLLM Worker

```shell
python3 -m dynamo.vllm \
  --model nvidia/Llama-3.3-70B-Instruct-FP8 \
  --tensor-parallel-size 8 \
  --enable-prefix-caching \
  --enable-chunked-prefill \
  --quantization fp8 \
  --kv-cache-dtype fp8 \
  --trust-remote-code \
  --enable-log-requests
```

The Core Idea: KV-Aware Routing

For each incoming request and each worker, NVIDIA Dynamo estimates a cost:

```
cost = overlap_score_weight * prefill_blocks + decode_blocks
```

Where:

- **prefill_blocks:** How much new KV must be computed (cache miss cost)
- **decode_blocks:** How busy the worker is on the decode side
- **overlap_score_weight:** Trades off TTFT vs decode latency

This turns routing from a **blind load balancer** into a **state-aware scheduler**.

Tuning

Up until this point, we've talked about why NVIDIA Dynamo + vLLM works: KV-aware routing, disaggregated prefill/decode, and cluster-level scheduling. But in practice, none of that matters unless the right knobs are turned.

We'll break this down using Workato's results and focus on three key metrics:

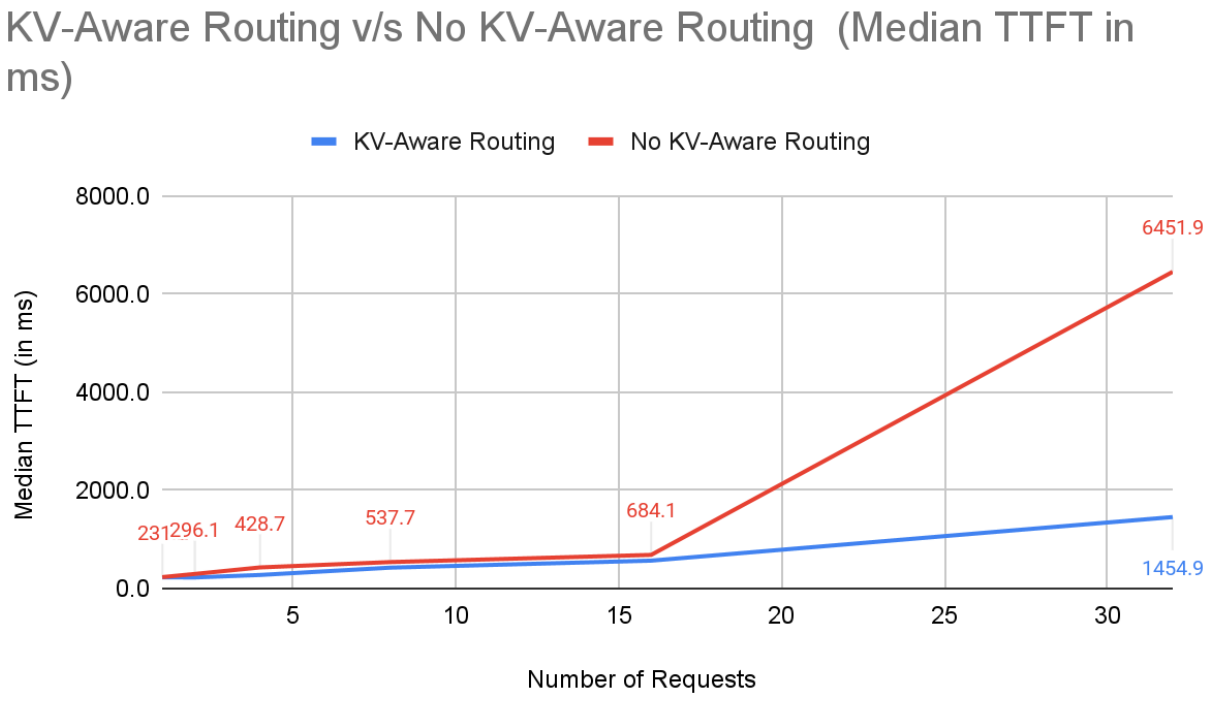
- **TTFT (Time-to-First-Token)**: How quickly the first token is produced after request arrival.
- **TPOT / ITL (Time-Per-Output-Token / Inter-Token Latency)**: How fast subsequent tokens are generated.
- **Throughput (QPS / Tokens/sec per GPU)**: Sustained work the system can handle.

TTFT: Prefill Cost & KV Reuse

Time-to-first-token (TTFT) is dominated by prefill work, the time the model spends processing the prompt to build KV states. Here, the tuning knobs are:

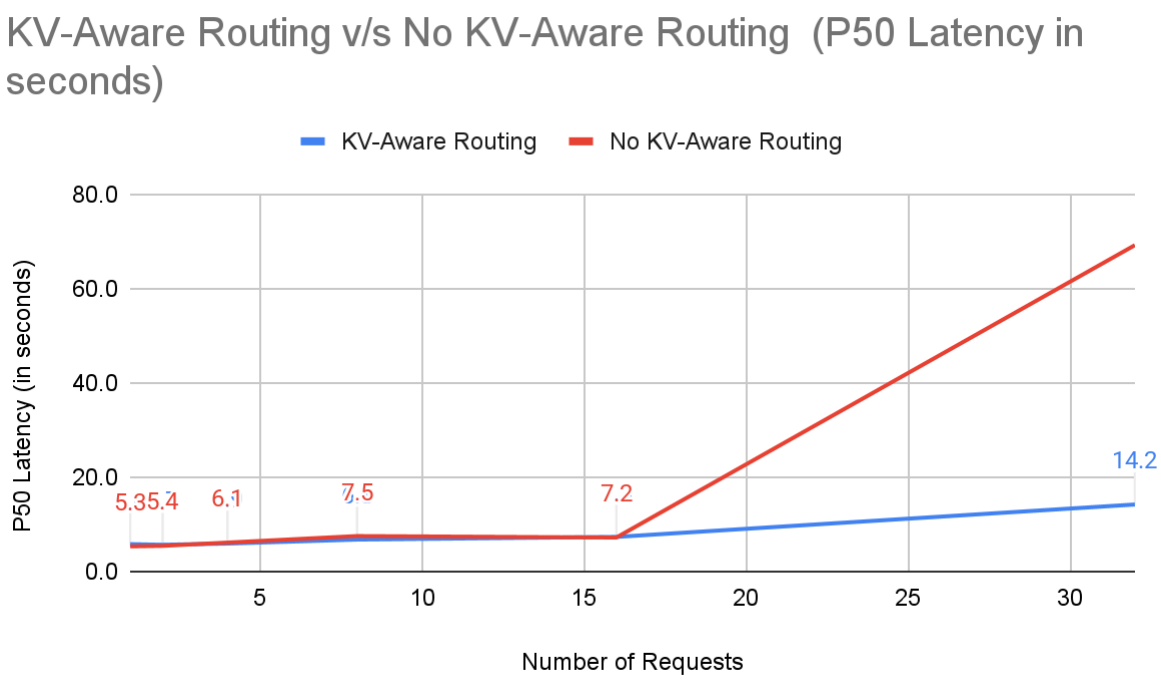
- `--enable-prefix-caching`: Reuses KV blocks for shared prefixes.
- `--enable-chunked-prefill`: Breaks prefill into manageable chunks to avoid GPU stalls.
- `--kv-cache-dtype fp8`: Reduces memory footprint and pressure, improving cache residency.

Why NVIDIA Dynamo excels:
Dynamo routes requests to the worker with most of the KV prefix already cached, avoiding redundant prefill. Even under high concurrency (32 prompts), the TTFT advantage is clear.



Median TTFT(ms) (2-node, 16–32 prompts):

Number of Requests	KV-Aware Routing with NVIDIA Dynamo	No KV-Aware Routing	% Improvement
16	566.7	684.1	**+17.2%**
32	**1454.9**	**6451.9**	**+77.5%**

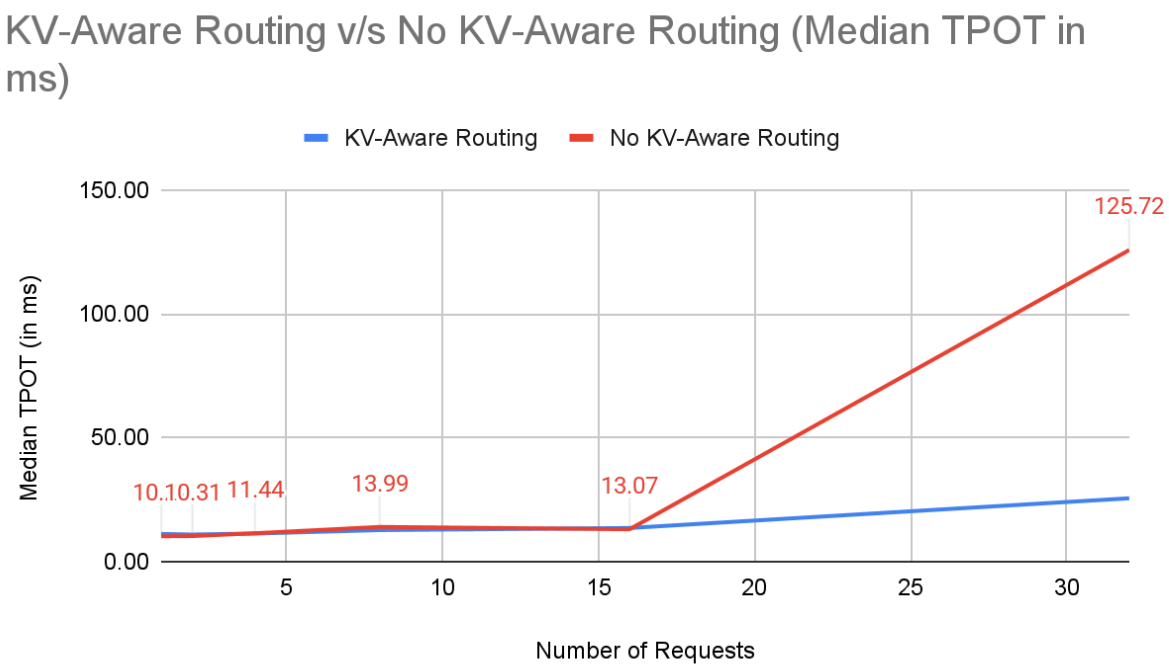


P50 Latency(s) (2-node, 16–32 prompts):

Number of Requests	KV-Aware Routing with NVIDIA Dynamo	No KV-Aware Routing	% Improvement
16	7.3	7.2	**−1.4%**
32	**14.2**	**69.2**	**+79.5%**

TPOT / ITL: Decode Load Balancing

TPOT and ITL depend on how busy the GPU is during decoding. Chunked prefill prevents one heavy request from stalling the decode pipeline. NVIDIA Dynamo further balances the decode load across workers, keeping TPOT stable even under bursts.



Data snapshot (2-node, 16–32 prompts):

Number of Requests	KV-Aware Routing with NVIDIA Dynamo	No KV-Aware Routing	% Improvement
16	13.57	13.07	-3.8%
32	**25.55**	**125.72**	**+79.97%**

Without prefix-aware routing, naive vLLM collapses under high concurrency because some workers get “prefill-heavy” requests while others idle, inflating TPOT and tail latency.

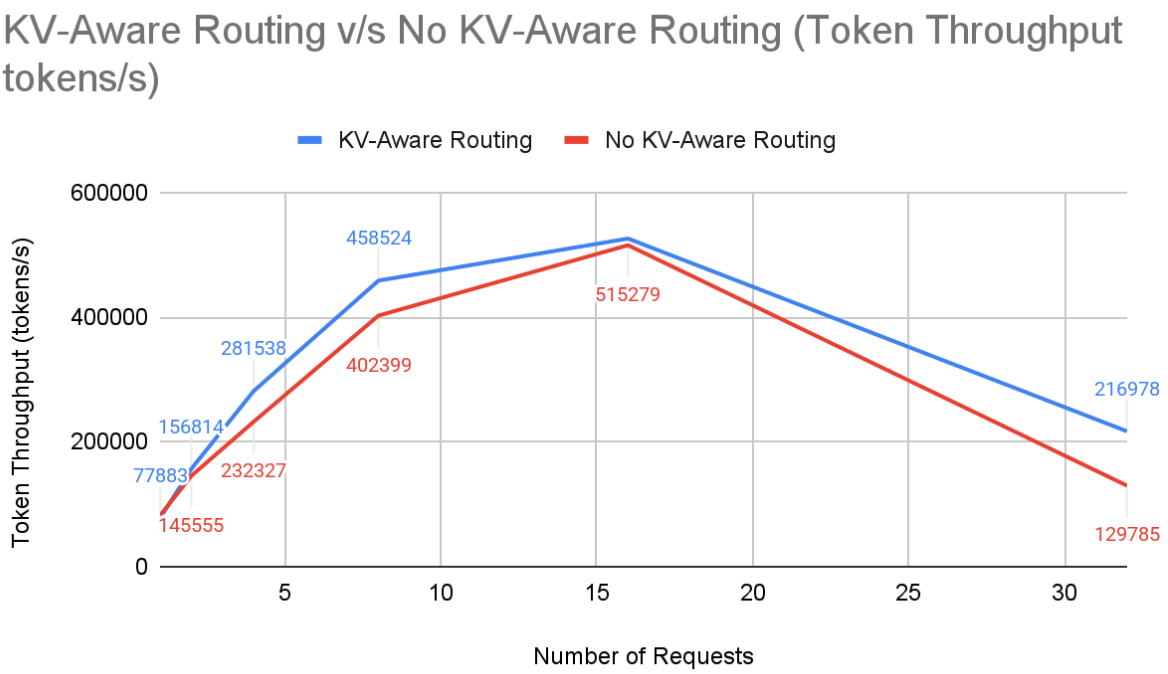
Where Dynamo wins:

- **KV reuse:** Avoids recomputing large prefill segments
- **Routing:** Sends requests to workers with hot KV caches

- **Scheduling**: Balances decode load, preventing hot GPU congestion

QPS & Token Throughput: GPU Utilization

Throughput per GPU at SLA (tokens/sec/GPU):



Data snapshot (2-node, 16–32 prompts):

Number of Requests	KV-Aware Routing with NVIDIA Dynamo	No KV-Aware Routing	% Improvement
16	525,898	515,279	+2.06%
32	**216,978**	**129,785**	**+67.19%**

With KV-aware routing, we achieved **13,561 tokens/sec per GPU versus 8,111 tokens/sec** without KV-aware routing, **a 67% uplift on identical hardware**. That improvement fundamentally changes capacity requirements. For a fixed SLA and token volume target, GPUs scale inversely with per-GPU throughput. A 1.67× throughput gain means each GPU now performs the work of 1.67 baseline GPUs, reducing required capacity to roughly 60% of the original footprint or 40% fewer GPUs for the same workload. In practical terms, a deployment that previously required 10 GPUs could now sustain the same load with approximately 6.

Fewer GPUs means fewer nodes, lower hourly burn, and less scaling overhead. Higher throughput per GPU doesn't just make the system faster: it reduces the number of GPUs needed to serve the same load, converting performance gains into measurable dollar savings.

Conclusion

Workato's results on DigitalOcean's Agentic Inference Cloud show that inference performance at scale is determined less by the model itself and more by how the system is architected around it.

For long-context, high-concurrency workloads, redundant prefill computation and uneven decode load quickly become the dominant cost and latency drivers. Simply adding GPUs does not address these inefficiencies. Coordinated routing and cache-aware scheduling do.

By deploying on NVIDIA Dynamo with vLLM on DOKS and enabling KV-aware routing, Workato eliminated redundant prefill work, improved load balance across workers, and stabilized latency under concurrency. On identical hardware, this translated to 67% higher tokens/sec per GPU, up to 79% lower end-to-end latency under load, and 77% lower TTFT — resulting in lower cost per token and fewer GPUs required to meet SLAs.

The key takeaway is architectural: inference efficiency is primarily a systems problem. When routing, cache management, GPU topology, and Kubernetes orchestration are coordinated, gains compound.

Depecated Technical Blog Draft

STO: Roxie Elliott
Contributors: Roxie Elliott Waverly Swinton Tayler Mehit Rithish Ramesh Tim Kim Karnik Modi
Go Live Date / Time: **March 11 at [TIME]**
Reminder: 2/16 is a US holiday

OUT OF DATE - use V2

Title: DigitalOcean delivers 67% Lower Inference Costs for Workato using NVIDIA Dynamo Architecture on H200 GPUs

Workato's AI Labs charter is to make AI-driven automation predictable, fast, and economically scalable across a wide range of enterprise workloads. For a platform processing millions of automated workflows, inference is core infrastructure and not a feature. Every millisecond of unnecessary latency and wasted GPU cycles has a direct cost.

The Workato team wanted answers to concrete questions:

- How much does cluster-level routing intelligence actually improve real-world throughput?
- What's the cost per token at production-grade SLAs?
- Does the complexity of a more sophisticated inference stack pay off compared to that of a well-tuned baseline?

To answer these questions, they chose DigitalOcean Kubernetes Service (DOKS) as the evaluation platform, utilizing the combination of managed Kubernetes simplicity, NVIDIA H200 GPU availability, performance optimization by DO, and infrastructure that didn't require them to become cloud specialists to operate.

How DigitalOcean Got Involved

DigitalOcean worked directly with Workato's AI Labs team to design, set up, and tune a distributed inference stack on DOKS. This wasn't a hand-the-docs-over engagement, but rather a white-glove experience where DigitalOcean's Solutions Architects team helped Workato configure NVIDIA Dynamo on top of DOKS, working through the architecture decisions, cluster topology options, and optimization knobs together.

The DigitalOcean team didn't just pick one configuration and measure it. We ran the evaluation across **1-node, 2-node,** and **4-node** DOKS clusters, comparing each topology for Dynamo + vLLM against an optimized vLLM baseline. This gave Workato a real map of the performance-to-cost trade-off space, not just a single data point.

The goal was to meet **production-grade SLA requirements** for Llama-3.3-70B (FP8), including high throughput, low latency, and cost efficiency. Key results from the evaluation:

- **Peak QPS:** 5.15 requests/sec on 2-node NVIDIA Dynamo + vLLM - **best-in-class across all tested configs**.
- **Throughput per GPU:** 13,561 tokens/sec on Dynamo 2-node vs. 8,111 tokens/sec for optimized vLLM on the same 2-node cluster: **67% higher** on identical hardware.
- **Cost efficiency:** $0.77 per 1M tokens on Dynamo 2-node: **27% lower** cost than optimized vLLM on the same 2-node cluster , and Dynamo 2-node achieves **67% lower** cost than vLLM running on a 4-node cluster using half the GPUs
- **Hardware efficiency: 33% lower hardware cost** using H200 vs. A100 for equivalent performance.
- **Latency:** Under high concurrency stress (conc=32), where optimized vLLM 2-node degrades to a 69s p50 and 6,452ms median TTFT, Dynamo holds at a **14.2s p50** and **1,455ms median TTFT**: a **79% lower end-to-end latency** and **77% lower time-to-first-token** on the same hardware under the same load.

These results exceeded the **goals Workato had for LLM inference**:

1. **High performance** for Llama-3.3-70B (FP8) in terms of throughput and latency.
2. **Advanced Dynamo optimizations** with KV cache-aware routing, chunked prefill, and cluster-level scheduling, layered on top of a high-performance vLLM backend.
3. **Cost efficiency, scalability, and SLA-grade reliability** for production workloads.

Scope disclaimer: This evaluation reflects a **POC environment** used to explore architectural trade-offs. These results represent Workato's evaluation setup only and do **not** reflect production workloads, recipes, or configurations.

The Model and Hardware

The Workato evaluation targeted **nvidia/Llama-3.3-70B-Instruct-FP8**, a production-grade frontier model that reflects the workloads Workato runs. The GPUs were NVIDIA H200s, chosen for their 141 GB HBM3e memory capacity and their ability to handle 100K-token context lengths without memory pressure forcing architectural compromises.

Parameter	Value
Model	nvidia/Llama-3.3-70B-Instruct-FP8
GPU	NVIDIA H200 (~141 GB HBM3e)
Tensor Parallelism	8 GPUs per worker node
Context Length	100,000 tokens

Cluster Topologies Tested	1-node, 2-node, 4-node
Platform	DigitalOcean Kubernetes Service (DOKS)

Memory footprint on H200:

- Model weights: ~70 GB (FP8, ~1 byte/parameter)
- KV cache (100K context, FP8): ~30-40 GB depending on paging strategy
- Activation buffers: ~10-15 GB during heavy prefill phases

With the H200's 141 GB HBM3e capacity, this all fits within a single GPU. However, the team used 8-way tensor parallelism per node to maximize sustained throughput and latency stability under a concurrent load.

How LLMs Process Requests and Why It Gets Expensive at Scale

Before getting into architecture decisions, it's worth understanding the mechanics that drive inference cost. Every LLM inference request goes through two phases.

- **Prefill** is where the model processes the entire input prompt and builds up internal "memory", technically called key/value (KV) states, for every token it has read. This phase is compute-heavy and scales linearly with prompt length. For long-context workloads (think 10K-100K token prompts), prefill can consume the majority of total inference cost.
- **Decode** is where the model generates tokens one at a time, using those cached KV states to predict each next token. This phase is latency-sensitive; users are waiting for each token to appear.

If two requests share a long common prefix, and they land on different nodes, both nodes recompute the same prefill independently. Neither knows what the other has already built. This is critical as at large token lengths and concurrencies, that redundancy compounds fast and becomes the dominant cost driver.

The solution is cluster-level routing.

Why KV Cache-Aware Routing Matters

When an LLM processes a prompt, it builds up internal key/value (KV) states that represent everything it has already “thought through” so far. If a new request shares a long prefix with a

previous one, the fastest path is not to recompute that reasoning, but to continue from where the model already left off.

KV cache-aware routing treats these partial reasoning states as reusable artifacts. Instead of sending requests to arbitrary GPUs, you route them to the GPU that already holds the relevant context. The benefits are simple but dramatic:

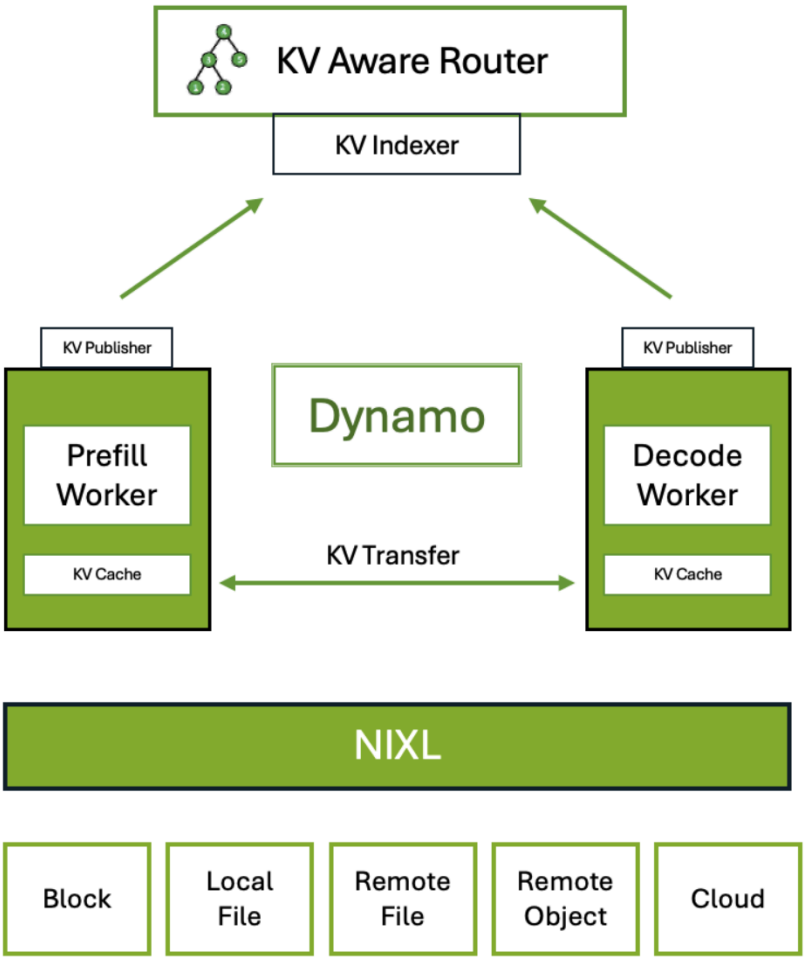
- Less wasted prefill
- Faster time-to-first-token (TTFT)
- More stable decode latency under load

That idea of "*route work to where the model already remembers things*" is the core systems insight behind NVIDIA Dynamo.

What is NVIDIA Dynamo?

NVIDIA Dynamo is an open-source, low-latency, modular inference framework designed to operate **above** individual inference engines. It is **engine-agnostic** and can orchestrate backends like **vLLM, TensorRT-LLM,** and **SGLang**.

In this evaluation, we use **vLLM** because it currently offers the **broadest feature coverage** and the cleanest integration surface.



Practically, Dynamo provides four things that a vanilla multi-node setup does not:

1. **A global scheduler** that sees all GPUs, not just one worker at a time
2. **A KV-aware router** that routes requests based on cache overlap, not round-robin
3. **A fast data plane (NIXL)** to move or reference KV state when needed
4. **A cluster-level KV cache manager** to track what lives where and when it should be reused or evicted

Smart Router (KV Cache-Aware)

The router is Dynamo's defining feature. It does **LLM-aware request routing** based on:

- Which worker already holds **overlapping KV cache blocks**
- The **tradeoff** between minimizing **prefill** cost (TTFT) and **decode** cost (ITL/TPOT)
- A **cost function** that scores candidate workers instead of blind round-robin

This is not HTTP-level load balancing. It is **token-level, state-aware scheduling**.

KV Cache Manager

The KV cache manager tracks:

- Which tokens live on which worker
- Which blocks are hot, cold, or evictable
- When to **reuse**, **offload**, or **recompute** cache segments

This enables **KV cache offloading**, **reuse across requests**, and **routing decisions** that are globally optimal instead of locally greedy.

The important point: Dynamo is not about making a single GPU faster. It is about **preventing the cluster from doing redundant work** and **keeping the right GPUs busy with the right phase of inference.**

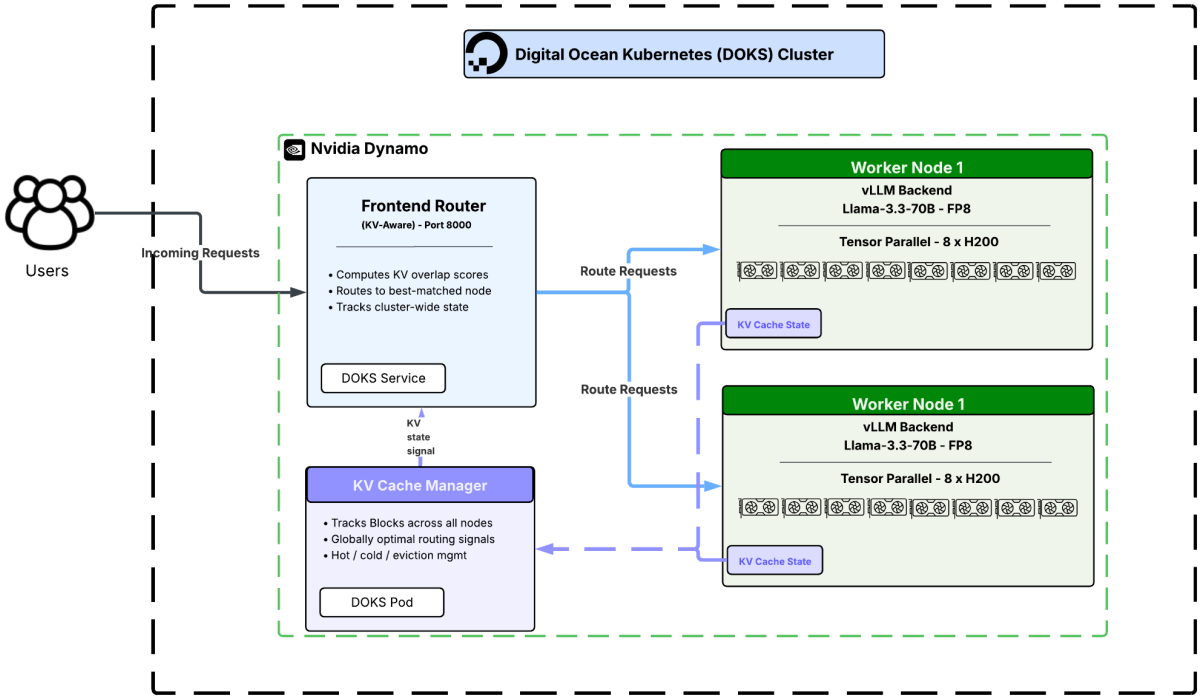
Workato's Architecture on DOKS

Deploying a cluster-aware inference stack like Nvidia Dynamo introduces real operational complexity. You're no longer running a few vLLM servers behind a load balancer; you're running a distributed system with a routing frontend, instrumented worker backends, a KV cache manager, and live coordination between all of them. Getting that right requires both the right framework and platform powering it all.

DOKS is what made this operationally viable for Workato. DigitalOcean's managed Kubernetes service handled the infrastructure orchestration, which allows the team to focus on inference architecture rather than cluster bootstrapping. Specifically:

- **GPU node scheduling** with NVIDIA device plugin support ensured inference pods always landed on the right hardware without manual intervention.
- **Independent replica management** for the routing frontend and worker backends meant the team could scale each layer separately as they tuned configurations.
- **Kubernetes-native service discovery** lets the router frontend find and route to worker pods cleanly, without custom networking work.
- **Operational simplicity**: DOKS abstracted away the control plane entirely, so Workato's engineers spent their time on inference problems, not orchestration problems.

The full cluster architecture for the 2-node deployment looked like this:



The frontend router of Nvidia Dynamo is a lightweight DOKS Service sitting in front of the GPU worker pods. The workers are full vLLM instances instrumented with Dynamo agents that continuously publish KV cache state back to the router and the cache manager. The KV cache manager is a separate pod that maintains the global view, tracking what's hot on which node, signaling evictions, and feeding the router's scoring function with fresh state.

Separating the router, workes, and cache manager as distinct DOKS workloads is what gave Workato the flexibility to tune each layer independently and make the architecture straightforward to scale horizontally as workload demands grow.

The Three Configurations Tested

Configuration 1: Naive vLLM (Baseline)

Multiple vLLM servers behind a simple round-robin load balancer. No prefix caching, no chunked prefill, no cluster-level awareness.

bash

```
Shell
vllm serve nvidia/Llama-3.3-70B-Instruct-FP8 \
```

```
  --tensor-parallel-size 8
```

Under long-context load, this configuration degrades predictably: cache hits are accidental, prefill work is duplicated across servers, and tail latency explodes as concurrency increases.

Configuration 2: Optimized vLLM (No Cluster-Level Routing)

The fair baseline: same model, same GPU, with vLLM's own optimizations fully enabled: prefix caching, chunked prefill, FP8 weights and KV cache. Each node makes smart local decisions, but there is no global view of cache state across the cluster.

```shell
vllm serve nvidia/Llama-3.3-70B-Instruct-FP8 \
  --tensor-parallel-size 8 \
  --enable-prefix-caching \
  --enable-chunked-prefill \
  --quantization fp8 \
  --kv-cache-dtype fp8 \
  --trust-remote-code
```

With this configuration, what actually happens under a 100k-token prefill load?

- **Repeated prefill computation:** If two prompts share a long prefix, but land on different workers, both workers recompute the same prefill.
- **Cache locality is accidental:** Whether a request benefits from warm KV cache is pure luck.
- **Load imbalance emerges:** Some workers get long prefills, others get decode-heavy traffic, queues diverge.
- **Tail latency explodes:** At higher concurrency, one unlucky routing decision can pin a worker with multiple heavy prefills while others idle.

Configuration 3: Dynamo + vLLM (KV-Aware Cluster Routing)

This is where the architecture fundamentally changes and out of the three configurations we tested, this was the clear winner.

At a high level, the Dynamo deployment splits the system into:

- A **Frontend** (the router + scheduler)
- A set of **Workers** (vLLM instances instrumented to talk to Dynamo)

You can think of this like a **smart warehouse dispatcher**:

- Naive routing is like sending every order to the shortest queue, even if another warehouse already has the item on the shelf.
- Dynamo is like checking **which warehouse already has most of the order pre-packed**, and routing the request there to avoid redoing work.

Dynamo Frontend (KV Router Enabled)

```
None
python3 -m dynamo.frontend --http-port 8000 --router-mode kv
```

The Dynamo frontend:

- Computes **KV overlap scores** against workers
- Applies the **routing cost function**
- Decides **where prefill and decode should execute**
- Tracks cluster-wide load and cache state

In this POC, we run the frontend with **KV routing enabled**.

Workers (deployed as DOKS pods on GPU nodes):

Each worker is a vLLM instance launched via Dynamo's wrapper. The integration exposes:

- **KV cache events:** What blocks are created, reused, or evicted
- **Metrics:** Queue depth, decode load, memory pressure
- **NIXL hooks:** For moving or referencing KV blocks across workers

Recipe: Dynamo + vLLM Worker

```
Shell
python3 -m dynamo.vllm \
  --model nvidia/Llama-3.3-70B-Instruct-FP8 \
  --tensor-parallel-size 8 \
  --enable-prefix-caching \
  --enable-chunked-prefill \
  --quantization fp8 \
```

```
  --kv-cache-dtype fp8 \
  --trust-remote-code \
  --enable-log-requests
```

The Core Idea: KV Cache-Aware Routing

For each incoming request and each worker, Dynamo estimates a cost:

```
None
cost = overlap_score_weight * prefill_blocks + decode_blocks
```

Where:

- **prefill_blocks:** How much new KV must be computed (cache miss cost)
- **decode_blocks:** How busy the worker is on the decode side
- **overlap_score_weight:** Trades off TTFT vs decode latency

This turns routing from a **blind load balancer** into a **state-aware scheduler**.

Configuration & Tuning Knobs

(What Actually Moves TTFT, TPOT, and Throughput)

We''ve talked about why Dynamo + vLLM works: KV-aware routing, disaggregated prefill/decode, and cluster-level scheduling. But in practice, none of that matters unless the right knobs are turned.

We'll break this down using Workato's POC results across naive vLLM, optimized vLLM, and Dynamo + vLLM.

We'll focus on three key metrics:

- **TTFT (Time-to-First-Token)**: How quickly the first token is produced after request arrival.
- **TPOT / ITL (Time-Per-Output-Token / Inter-Token Latency)**: How fast subsequent tokens are generated.
- **Throughput (QPS / Tokens/sec per GPU)**: Sustained work the system can handle.

Cluster-Level Scheduling (What Dynamo Adds)

Knobs:

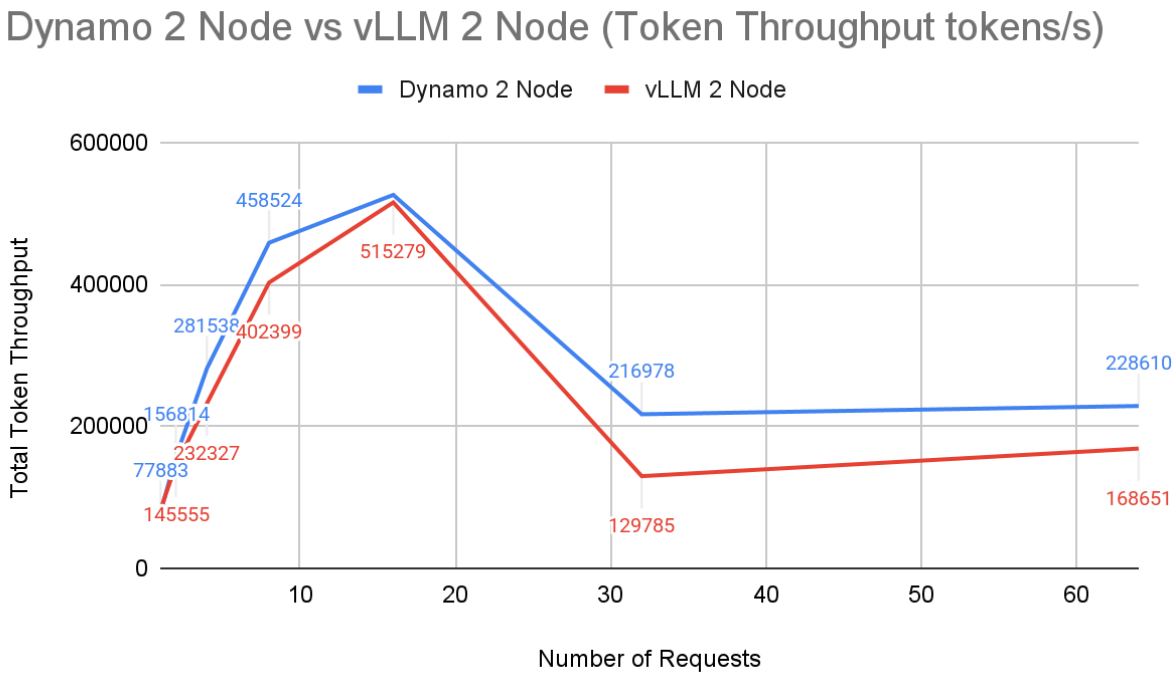
```shell
Frontend: --router-mode kv

python3 -m dynamo.frontend --http-port 8000 --router-mode kv
```

- Enables KV cache-aware routing.
- **Why it matters:**
  - Requests are sent to workers that already have the prefix cached instead of just the shortest queue.

This is the **single biggest system-level difference** vs optimized vLLM.

QPS & Token Throughput: GPU Utilization

Throughput per GPU at SLA (tokens/sec/GPU):



Data snapshot (16 GPUs, 16–64 prompts, Dynamo vs vLLM):

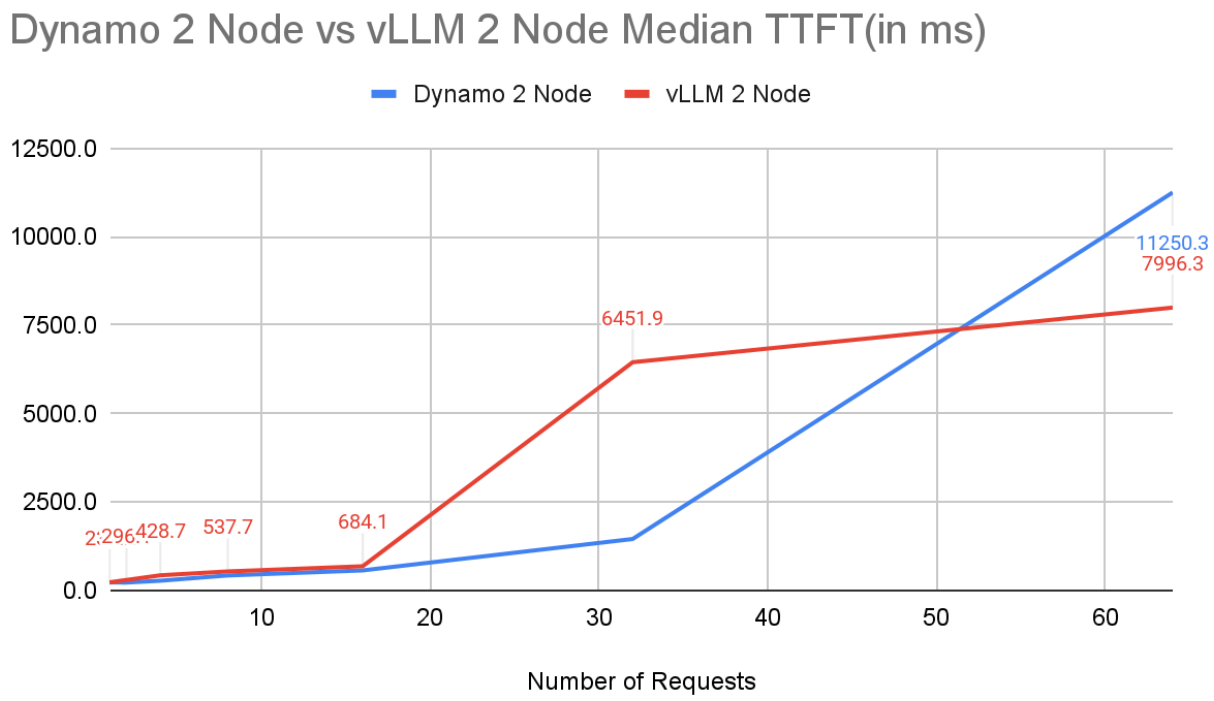
Concurrency	Optimized vLLM Total Tokens/sec	Dynamo Total Tokens/sec	Optimized vLLM Tokens/sec/GPU	Dynamo Tokens/sec/GPU	% Improvement
16 prompts	515,279	525,898	32,205	32,868	2.06%
32 prompts	129,785	216,978	8,111	13,561	67.19%
64 prompts	168,465	228,704	10,529	14,294	35.76%

TTFT: Prefill Cost & KV Reuse

Time-to-first-token (TTFT) is dominated by prefill work, the time the model spends processing the prompt to build KV states. Here, the tuning knobs are:

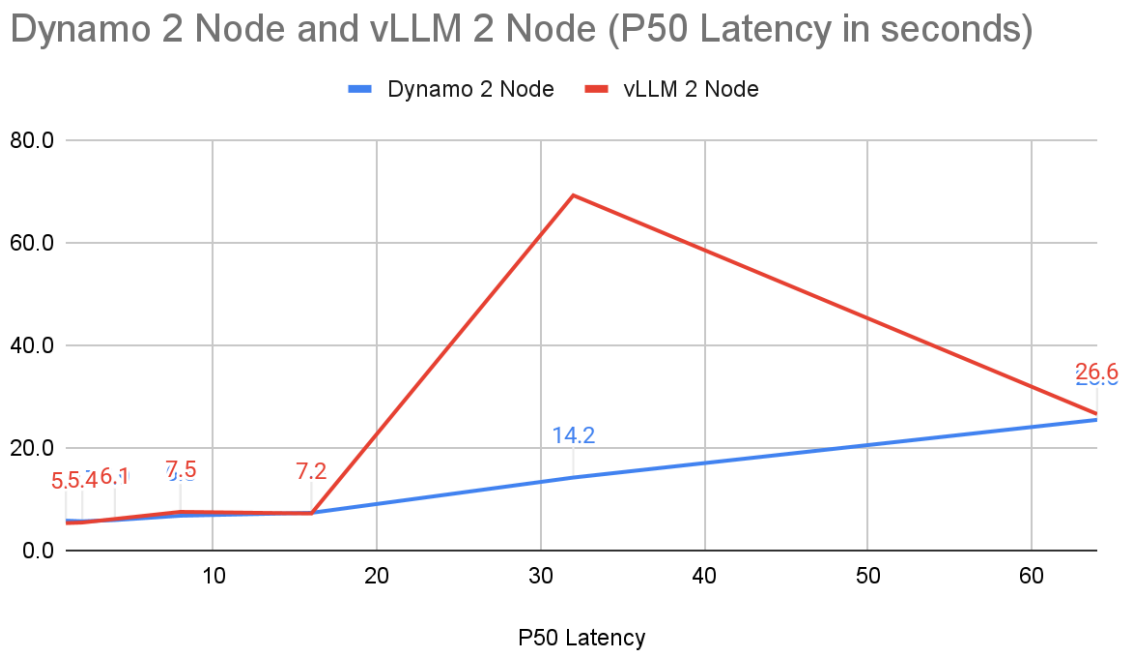
- `--enable-prefix-caching`: Reuses KV blocks for shared prefixes.
- `--enable-chunked-prefill`: Breaks prefill into manageable chunks to avoid GPU stalls.
- `--kv-cache-dtype fp8`: Reduces memory footprint and pressure, improving cache residency.

Why Dynamo excels:
Dynamo routes requests to the worker with most of the KV prefix already cached, avoiding redundant prefill. Even under high concurrency (32 prompts), the TTFT advantage is clear.



Median TTFT(ms) (2-node, 16–32 prompts, Dynamo vs vLLM):

Number of Prompts	Dynamo	vLLM	% Improvement
16	566.7	684.1	**+17.2%**
32	1454.9	6451.9	**+77.5%**
64	11250.3	7996.3	**−40.7%**

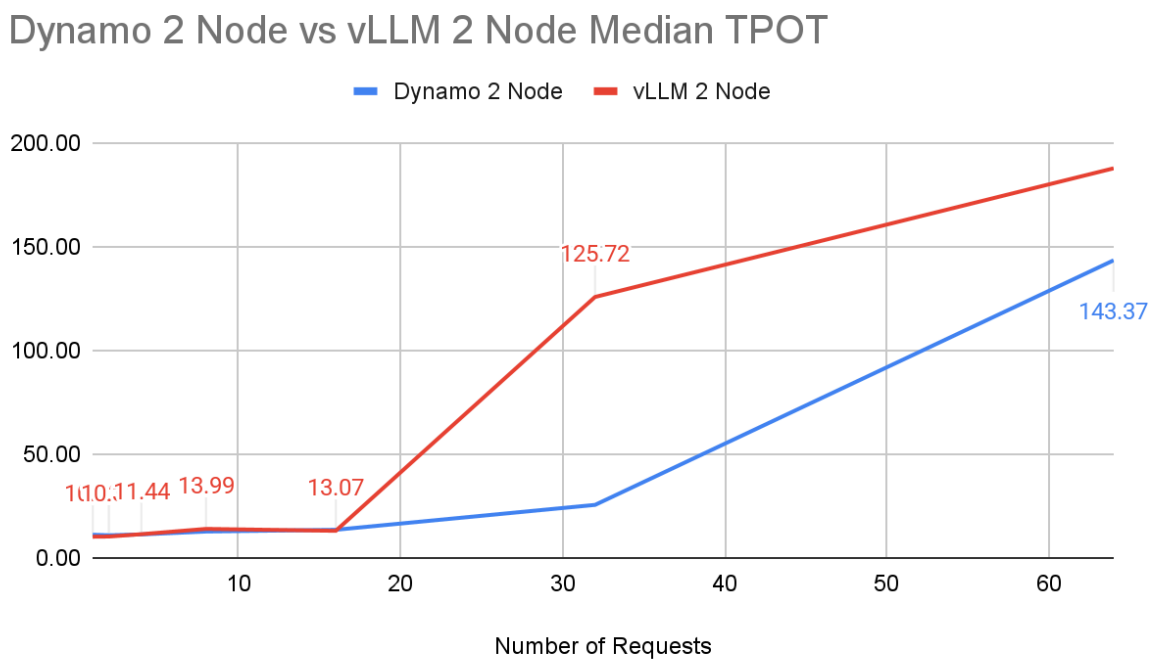


P50 Latency(s) (2-node, 16–64 prompts, Dynamo vs vLLM):

Number of Prompts	Dynamo	vLLM	% Improvement
16	7.3	7.2	**−1.4%**
32	14.2	69.2	**+79.5%**
64	25.5	26.6	**+4.1%**

TPOT / ITL: Decode Load Balancing

Time per output token (TPOT) / inter-token latency (ITL) depends on how busy the GPU is during decoding. Chunked prefill prevents one heavy request from stalling the decode pipeline. Dynamo further balances the decode load across workers, keeping TPOT stable even under bursts.



Data snapshot (2-node, 32 prompts):

Prompts	Dynamo (ms)	vLLM (ms)	Improvement %
16	13.57	13.07	-3.8%
32	**25.55**	**125.72**	**+79.97%**
64	143.37	187.69	+23.36%

Without KV-aware routing, naive vLLM collapses under high concurrency because some workers get “prefill-heavy” requests while others idle, inflating TPOT and tail latency.

Where Dynamo wins:

- **KV reuse:** Avoids recomputing large prefill segments
- **Routing:** Sends requests to workers with hot KV caches
- **Scheduling:** Balances decode load, preventing hot GPU congestion

Naive vLLM is like 16 cooks in separate kitchens, each starting from scratch every time. Dynamo is like a smart head chef: directs orders to kitchens that already have ingredients

prepped, shares tasks, and prevents bottlenecks. The result: more meals ready faster and with fewer burnt dishes.

What's Next

Even with these results, there's room to improve. Future work in this space includes smarter KV cache eviction policies, more aggressive prefill/decode disaggregation, pre-warming of hot prefix segments, and expanded quantization options for memory efficiency. As inference frameworks mature and GPU infrastructure continues to improve, the gap between naive and optimized deployments will only widen making it more important to get the architecture right early.

Conclusion

Workato's evaluation on DigitalOcean demonstrates that the performance ceiling for LLM inference isn't set by the model, it's set by the system around it. By combining KV cache-aware routing, chunked prefill, FP8 precision, and cluster-level scheduling - all orchestrated with Dynamo architecture on DOKS, the team achieved 67% higher per-GPU throughput, 67% lower cost per Million token, and 77% better TTFT under high concurrency compared to a well-tuned baseline.

For any team building production AI services: the compute is necessary but not sufficient. The orchestration layer is where the economics of AI at scale get solved.

For platforms like Workato, this is more than a benchmark: it’s a **blueprint for predictable, high-throughput, cost-efficient AI services**. The next evolution of Dynamo promises even more performance wins, making it an essential tool for enterprise-scale LLM deployments.

Case Study

STO: Roxie Elliott
Contributors: Waverly Swinton Tayler Mehit
Go Live Date / Time: **March 3 at [TIME]**
Reminder: 2/16 is a US holiday

Reviewer / Activity	Approved?	Send	Deadline	Notes/Comments
Meghan Grady	In progress	2/11	2/13	
Laura Schaffer	Not started	2/17	2/19	
Legal (Ironclad signoff)	Approved	2/26	2/27	
Workato review	Not started	2/13	2/26	[EXTERNAL] Workato - Customer Story R…
NVIDIA approval	Not started	2/13	2/26	[EXTERNAL] NVIDIA - Workato Customer …

How Workato Is Scaling the Agentic Enterprise with Optimized NVIDIA H200 Clusters from DigitalOcean

Pull quote: *"DigitalOcean stood out to us through sheer speed and commitment. That level of trust and responsiveness was a deciding factor. The solutions team has been exceptional. With the hyperscalers, we'd probably still be waiting in a queue." —Oscar Wu, AI Research Scientist and Founder of AI Labs, Workato*

Workato is the leader in AI-driven enterprise automation, helping over 12,000 global customers unify data, processes, and applications. In 2025, the company launched Workato One, a unified platform that brings together everything organizations need to build and deploy enterprise-ready agents across business functions. At the heart of Workato's AI innovation is Workato AI Labs, a San Francisco-based research hub dedicated to advancing the science of autonomous agents through synthetic evaluation, automated reinforcement learning, and custom model optimization. To power this frontier research, Workato partnered with DigitalOcean to build their high-performance infrastructure using NVIDIA H200 GPU Droplets and DigitalOcean Managed Kubernetes, achieving a 2-3x acceleration in time-to-value and a 67% reduction in inference costs.

Workato's Inference and Support Needs

For the lean engineering team at Workato AI Labs, the primary obstacle wasn't just finding GPUs, but also the operational overhead required to manage them. As a research-focused unit, every hour spent debugging network configurations or container orchestration was an hour lost to core AI innovation. The team needed an infrastructure partner that functioned as an

extension of their own engineering staff, providing both rapid deployment and proactive optimization.

The AI Labs required infrastructure capable of handling distributed training and sustained, reasoning-heavy inference under real production load. Not only was DigitalOcean able to provide high-performance NVIDIA HGXTM H200 GPUs faster than any other provider for the team to begin their work, the Workato team quickly discovered the significant performance boost and TCO improvement as a result of DigitalOcean's inference-optimized architecture and simplified experience.

"Before DigitalOcean, we didn't have a dedicated solution for in-house training and multi-node serving, and that was a major blocker for AI research," said Oscar Wu, AI Research Scientist at Workato. "DigitalOcean was the fastest provider to get us up and running, enabling us to advance our AI programs. Beyond access to GPUs, the collaboration on performance optimization coupled with the support from their team of solutions architects, accelerated our progress by roughly two to three times."

Beyond procurement of GPUs, Workato was also hesitant about the lack of high-touch support from hyperscale providers. For a small team, the black box nature of legacy cloud support was a major risk. They required a partner who could provide deep technical root-cause analysis rather than generic documentation.

From Evaluation to Impact: A High-Performance Backbone for Frontier Research

To move beyond these blockers, Workato conducted a rigorous evaluation of multiple providers, including both hyperscale cloud providers such as AWS and Google Cloud Platform and AI neoclouds including Baseten. Workato ultimately selected DigitalOcean for its superior speed of execution and the specialized expertise of its engineering team, who worked hand-in-hand with Workato to optimize GPU performance and costs. This partnership allowed Workato to build a dedicated infrastructure that could handle both intensive model training and high-throughput inference with a level of agility the hyperscalers couldn't match.

The transition to an optimized DigitalOcean environment fundamentally transformed Workato's research economics. By controlling the full hardware stack—centered on interconnected NVIDIA H200 GPU clusters—the Labs team was able to reduce the cost of serving specialized agents while dramatically increasing the speed at which those agents respond to business events. These H200 clusters provide a 33% hardware cost advantage over the A100 series while delivering significantly higher memory bandwidth, a critical factor for Workato's high-stakes agentic workloads.

Precision Engineering and Quantified Results

To support frontier models like Llama-3.3-70B and Llama-3.1-8B, the Labs team implemented a highly tuned software and hardware stack. By serving models in FP8 and FP4 numerical precision, they maximized throughput without sacrificing accuracy. The team utilized NVIDIA Dynamo v0.4.1 combined with vLLM (virtual Large Language Model), leveraging PagedAttention to manage memory for long-context agentic reasoning. To facilitate multi-node communication for larger models, DigitalOcean configured the nodes with NVLink. As Huang explains, "We use them for just doing multi-node stuff, like training bigger models that are not trainable on one node."

This technical precision led to immediate, measurable gains:

- **67% Lower Cost per 1M Tokens:** Infrastructure costs dropped to just **$0.77 per million tokens**, significantly outperforming high-tier API providers.
- **67% Throughput Increase:** The optimized H200 setup reached **13,561 tokens/sec per GPU**, a massive leap from the baseline of 8,111 tokens/sec.
- **Reliable Low Latency:** For real-time "Genie" interactions, the team achieved a **P50 latency of 13 seconds** at 3.38 requests per second even on heavy 70B parameter models.

Deep Engineering Partnership as a Research Multiplier

Beyond raw metrics, the deciding factor for Workato was the deep AI engineering partnership that turned DigitalOcean into a true extension of the Workato team. During the critical GPU setup and testing phase, the team hit a roadblock with a broken etcd image reference in a standard Docker configuration. Rather than providing a generic support ticket response, DigitalOcean's Solutions Architect, Rithish, worked directly with Workato to identify the root cause and stabilize the cluster. This proactive intervention saved the team an estimated one week of troubleshooting.

This level of collaborative engineering, which encompasses setup, rigorous benchmarking, and ongoing maintenance, gave Workato the confidence to move at research speed. "DigitalOcean is the only provider that lets me focus on research instead of managing infrastructure," said Kevin Huang, Infrastructure Engineer at Workato AI Labs. "We can provision GPUs quickly, deploy inference workloads in production, and iterate on real customer traffic without getting bogged down in platform complexity. That speed has been critical to maintaining our momentum."

This collaborative synergy turned an aggressive sprint into a success. "Being able to collaborate on an aggressive sprint and achieve results that were impressive were a major factor in our decision to use DigitalOcean," says Wu. The result was a "two to three X speed up in time to value."

What's Next for Workato: Scaling the Autonomous Enterprise

With a stable infrastructure foundation, Workato AI Labs is now focused on the next generation of agentic capabilities. This includes developing Self-Improving Agents that can observe business operations, identify inefficiencies, and autonomously design new process "recipes."

To support this, Workato is scaling its Agent Knowledge Base—a continuously evolving memory layer—and its support for the Model Context Protocol (MCP), which allows agents to share skills across different platforms.

"We're building towards a future where agents can observe business operations, and optimize existing ones in this continuous self-improving loop," says Wu. "DigitalOcean and NVIDIA's infrastructure is foundational to that work, as we need reliable, high performance compute to keep on pushing the frontier."

[v3] - Social Posts

Social Posts

STO: Jakub Lukaszczyk / Julia Downing
Contributors: Waverly Swinton Tayler Mehit Jakub Lukaszczyk Julia Downing TBD
Go Live Date / Time: **March 3 at [TIME]**

DODX ticket
Social ticket

LIVE LINKS

- **LinkedIn**
- **X**
- **Threads**
- **Instagram**
- **Bluesky**

Reviewer / Activity	Approved?	Send	Deadline	Notes/Comments
Meghan Grady	In progress	2/12	2/19	
Laura Schaffer	In progress	2/18	2/19	
Workato review	Not started	2/19	2/26	[EXTERNAL] Workato - Customer Story R…
NVIDIA approval	Approved	2/19	2/26	[EXTERNAL] NVIDIA - Workato Customer …

In-asset Messaging

~~Framework~~	~~Option A (Technical + Story)~~	~~Framework~~	~~Option B (Meghan's Mashup)~~		Option C (Technical / Operator)
~~Hook~~	~~Workato processes 1 trillion automated workloads. Inference performance is a hard requirement.~~	~~Hook~~	~~Workato processes 1 trillion automated workloads. Inference performance is a hard requirement. Here's how DigitalOcean's Agentic Inference Cloud delivered.~~	Hook	Workato's AI Research Lab achieved 79% lower end-to-end latency and 67% lower model cost on DigitalOcean's Agentic inference Cloud.
~~Challenge~~	~~Delivering agentic AI across thousands of enterprise customers means 100K-token context lengths and high~~	~~Results~~	~~The results were clear. On identical hardware: Running Llama-3.3-70B-Instruct-FP8 on NVIDIA GPU Droplets, we achieved:~~ ~~• 67% higher throughput per GPU~~ ~~• 79% lower end-to-end latency~~	What we tested	The setup: Llama-3.3-70B (FP8), 100K context, NVIDIA H200 GPU droplets 2-node / 16-GPU cluster on DigitalOcean Kubernetes (DOKS)

	~~concurrency. Adding GPUs alone doesn't solve it~~		~~• 77% faster time-to-first-token~~ ~~• 67% lower model cost using half the GPUs~~		
~~Results~~	~~The results on identical hardware:~~ ~~• 67% higher throughput per GPU~~ ~~• 79% lower end-to-end latency~~ ~~• 77% faster time-to-first-token~~ ~~• 67% lower model cost using half the GPUs~~	~~Solution~~	~~DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached.~~ ~~• NVIDIA Dynamo~~ ~~• vLLM~~ ~~• H200 GPUs on DigitalOcean Kubernetes OKS~~	What Worked	Inference-optimized architecture that works: NVIDIA Dynamo with KV-aware routing + vLLM on DOKS Cluster routes requests to where KV cache already exists → better utilization + lower tail latency.
~~Solution~~	~~DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached~~ ~~• NVIDIA Dynamo~~ ~~• vLLM~~ ~~• H200 GPUs on DOKS~~	~~How~~	~~At 100K token context, duplicate prefill is your biggest cost. Dynamo routes requests to the GPU that already has the context cached. Less recompute. Lower latency. Better economics.~~	Results / Proof points	The results: • 67% higher throughput per GPU • 79% lower end-to-end latency • 77% faster time-to-first-token • 67% lower model cost using half the GPUs
~~CTA~~	~~Read the Technical Blog for full benchmark + architecture breakdown~~	~~CTA~~	~~See how DigitalOcean helped Workato build inference infrastructure that scales with enterprise AI.~~	Tagline	DigitalOcean's Agentic Inference Cloud: Built for inference at scale

Social Post-Copy

Blog link:
https://www.digitalocean.com/blog/workato-nvidia-technical-deep-dive-agentic-inference-cloud

LinkedIn
Workato processes 1 trillion automated workloads. Inference performance isn't optional. It's a hard requirement.

DigitalOcean's Agentic Inference Cloud powered by NVIDIA Hopper GPUs delivered significant results:
✅ 67% higher throughput per GPU
✅ 79% lower end-to-end latency
✅ 77% faster time-to-first-token
✅ 67% lower model cost using half the GPUs

NVIDIA Dynamo with KV-aware routing sends every request to where the context already lives. Less recompute. Lower latency. Better economics. 🌊

DigitalOcean
Mar 3

Workato processes 1 trillion automated workloads. Inference performance isn't optional. It's a hard requirement.

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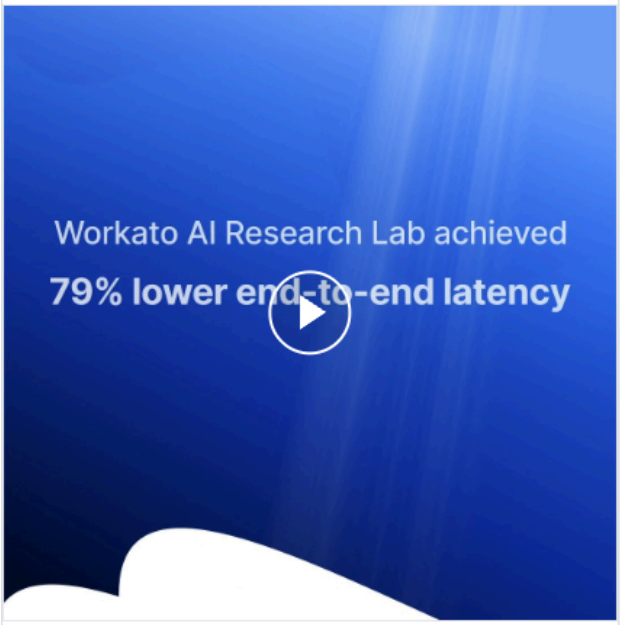
NVIDIA Dynamo with KV-aware routing sends every request to where the context already lives. Less recompute. Lower latency. Better economics. 🌊

Show Less



Like Comment Share

DigitalOcean Author

Learn more. 🔗

Like | Reply

X

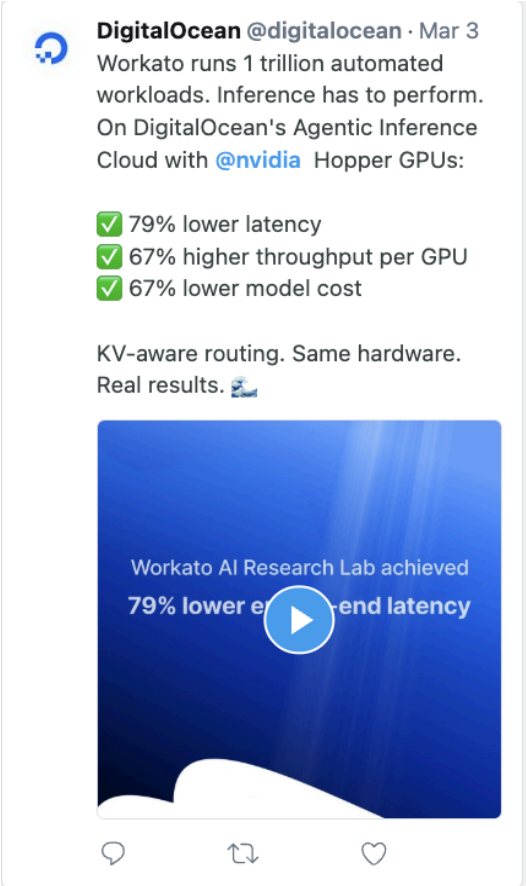
Workato runs 1 trillion automated workloads. Inference has to perform. On DigitalOcean's Agentic Inference Cloud with @nvidia Hopper GPUs:

✅ 79% lower latency

✅ 67% higher throughput per GPU

✅ 67% lower model cost

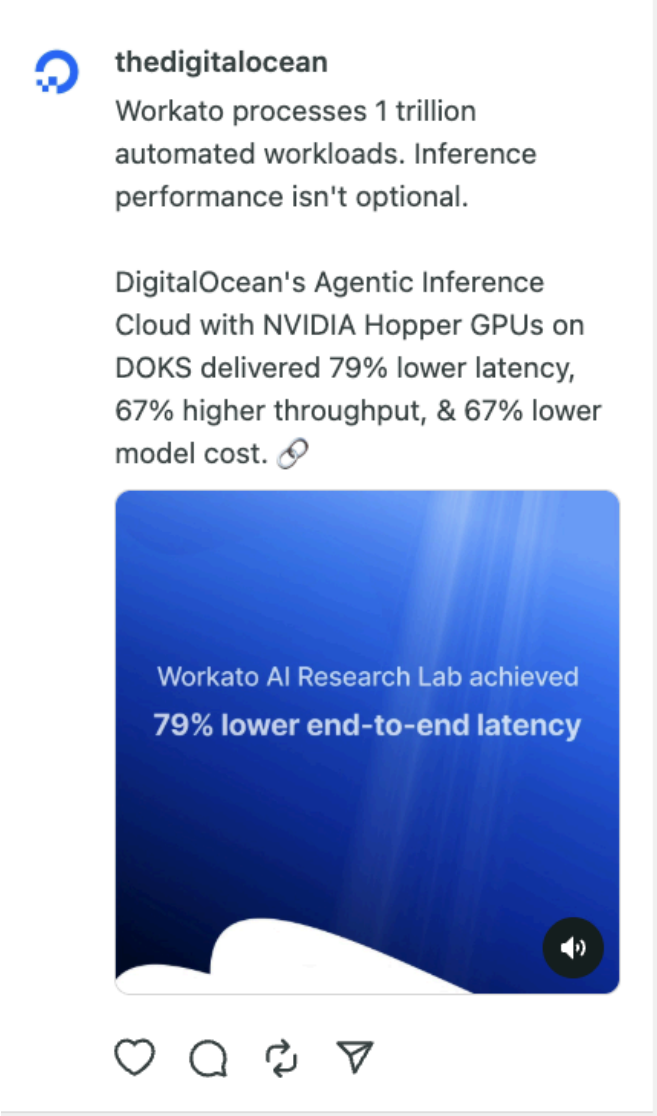
KV-aware routing. Same hardware. Real results. 🌊



Threads

Workato processes 1 trillion automated workloads. Inference performance isn't optional.

DigitalOcean's Agentic Inference Cloud with NVIDIA Hopper GPUs on DOKS delivered 79% lower latency, 67% higher throughput, & 67% lower model cost. 🔗



Instagram

One trillion automated workloads. One infrastructure partner. 🤝

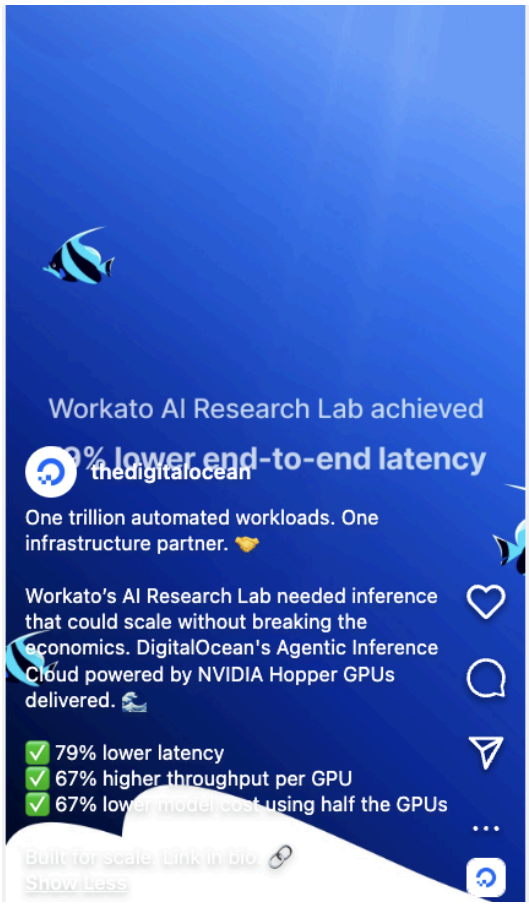
Workato's AI Research Lab needed inference that could scale without breaking the economics. DigitalOcean's Agentic Inference Cloud powered by NVIDIA Hopper GPUs delivered. 🌊

✅ 79% lower latency

✅ 67% higher throughput per GPU

✅ 67% lower model cost using half the GPUs

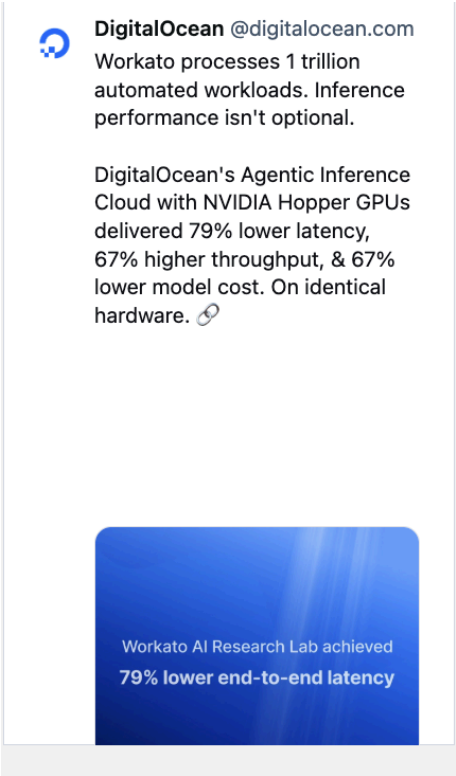
Built for scale. Link in bio. 🔗



Bluesky

Workato processes 1 trillion automated workloads. Inference performance isn't optional.

DigitalOcean's Agentic Inference Cloud with NVIDIA Hopper GPUs delivered 79% lower latency, 67% higher throughput, & 67% lower model cost. On identical hardware. 🔗



○

Social Posts Draft

Deprecated - see v3

Animation

Option A: Current version updated to the [v2] Technical Blog.

Intro: [Intro Logo “Workato Powered by DigitalOcean x NVIDIA]

Frame 1: *The Hook*
~~Workato's AI Labs had one question: does a smarter inference stack actually pay off?~~
Workato processes 1 trillion automated workloads. They needed GPUs that could keep up. Here's what DigitalOcean delivered.

Frame 2: *Results*
Same hardware. Same model
67% higher throughput per GPU
77% faster time-to-first-token
67% lower model cost using half the GPUs

Frame 3: *Setup*
Model: Llama-3.3-70B (FP8)
Context: 100K tokens
Hardware: NVIDIA H200s
Platform: DigitalOcean Kubernetes (DOKS)
Test: NVIDIA Dynamo + vLLM vs. optimized vLLM baseline

~~**Frame 3:** *Results*~~
~~Same hardware. Same model~~
~~67% higher throughput per GPU~~
~~77% faster time-to-first-token~~
~~67% lower model cost using half the GPUs~~

Frame 4: *Why*
At 100K-token context, duplicate prefill is your biggest cost
Dynamo routes requests to the GPU that already has the context cached
Less recompute. Lower latency. Better economics

Frame 5: *CTA*
Read the Technical Blog for full benchmark + architecture breakdown

Option B: New version converging [v2] Technical Blog and “Social Posts Draft”.

Intro: [Intro Logo “Workato Powered by DigitalOcean x NVIDIA]

Frame 1: *The Hook*
Workato processes 1 trillion automated workloads. ~~Inference infrastructure has to keep up~~
Inference performance is a hard requirement.

Frame 2: *Challenge*
Delivering agentic AI across thousands of enterprise customers means 100K-token context lengths and high concurrency. Adding GPUs alone doesn't solve it

Frame 3: Results
The results on identical hardware:

- 67% higher throughput per GPU
- 79% lower end-to-end latency
- 77% faster time-to-first-token
- 67% lower model cost using half the GPUs

Frame 4: *Solution*
DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached

- NVIDIA Dynamo
- vLLM
- H200 GPUs on DOKS

~~**Frame 4:** *Results*~~
~~The results on identical hardware:~~

- ~~67% higher throughput per GPU~~
- ~~79% lower end-to-end latency~~
- ~~77% faster time-to-first-token~~
- ~~67% lower model cost using half the GPUs~~

Frame 5: *CTA*

See how DigitalOcean helped Workato build inference infrastructure that scales with enterprise AI.

Option C: Requested mash-up of Option A & B.

Frame 1: *The Hook*
Workato processes 1 trillion automated workloads. Here's what DigitalOcean delivered.

Frame 2: *Results*
The results were clear. On identical hardware:
• 67% higher throughput per GPU
• 79% lower end-to-end latency
• 77% faster time-to-first-token
• 67% lower model cost using half the GPUs

Frame 3: *Solution*
DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached.
• NVIDIA Dynamo
• vLLM
• H200 GPUs on DOKS

Frame 4: *How*
At 100K-token context, duplicate prefill is your biggest cost. Dynamo routes requests to the GPU that already has the context cached. Less recompute. Lower latency. Better economics.

Frame 5: *CTA*
See how DigitalOcean helped Workato build inference infrastructure that scales with enterprise AI.

Framework	Option A (Technical)	Framework	Option B (Technical + Story)	Framework	Option C (Meghan's Mashup)
Hook	Workato processes 1 trillion automated workloads. They needed GPUs that could keep up. Here's what DigitalOcean delivered.	Hook	Workato processes 1 trillion automated workloads. Inference performance is a hard requirement.	Hook	Workato processes 1 trillion automated workloads. performance is a hard requirement. Here's how DigitalOcean delivered.

Results	Same hardware. Same model 67% higher throughput per GPU 77% faster time-to-first-token 67% lower model cost using half the GPUs	Challenge	Delivering agentic AI across thousands of enterprise customers means 100K-token context lengths and high concurrency. Adding GPUs alone doesn't solve it	Results	The results were clear. On identical hardware: • 67% higher throughput per GPU • 79% lower end-to-end latency • 77% faster time-to-first-token • 67% lower model cost using half the GPUs
Setup	Model: Llama-3.3-70B (FP8) Context: 100K tokens Hardware: NVIDIA H200s Platform: DigitalOcean Kubernetes (DOKS) Test: NVIDIA Dynamo + vLLM vs. optimized vLLM baseline	Results	The results on identical hardware: • 67% higher throughput per GPU • 79% lower end-to-end latency • 77% faster time-to-first-token • 67% lower model cost using half the GPUs	Solution	DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached. • NVIDIA Dynamo • vLLM • H200 GPUs on DOKS
Why	At 100K-token context, duplicate prefill is your biggest cost Dynamo routes requests to the GPU that already has the context cached. Less recompute. Lower latency. Better economics	Solution	DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached • NVIDIA Dynamo • vLLM • H200 GPUs on DOKS	How	At 100K-token context, duplicate prefill is your biggest cost. Dynamo routes requests to the GPU that already has the context cached. Less recompute. Lower latency. Better economics.
CTA	Read the Technical Blog for full benchmark + architecture breakdown	CTA	See how DigitalOcean helped Workato build inference infrastructure that scales with enterprise AI.	CTA	See how DigitalOcean helped Workato build inference infrastructure that scales with enterprise AI.

Frame	**Message**	**Copy**	
1	Workato is doing super cool stuff powered by DO	Workato One enables enterprise agentic AI, deploying over 1 trillion tasks (a day??), now	Workato’s AI Research Labs achieved

	Workato delivers at enterprise scale: 1 Trillion tasks for thousands of customers. The engine? DigitalOcean's vertically integrated Inference Cloud.	powered by DigitalOcean’s Agentic Inference Cloud. 1 Trillion Tasks. Workato One's enterprise AI is scaling the future with DigitalOcean's Agentic Inference Cloud.	67% Higher Throughput 77% Faster Time-to-First-Token 67% Lower Inference Costs DigitalOcean Agentic Inference Cloud x NVIDIA Hopper GPUs
2	By using DO’s vertically integrated inference platform, they get by better results - Technical proof point	On DigitalOcean’s vertically integrated inference cloud, Workato AI Labs experienced: 67% Higher Throughput 77% Faster Time-to-first-token 67% Lower Inference costs	
3	[Optional] - customer proof point that backs up the above stats ⇒ pull quote	The collaboration on performance optimization coupled with the support from their team of solutions architects, accelerated our progress by roughly two to three times.	
5	CTA?	Purpose-built for Production AI. Optimized NVIDIA clusters on DigitalOcean	

The hardest problem in AI is no longer training models.

It’s running inference at scale — under real demand, with predictable economics.

Agentic AI demands reliable, high-performance compute. Workato is powering their next-gen agents on interconnected #NVIDIA H200 clusters and DigitalOcean Managed Kubernetes.

Stop managing infra, start innovating. That’s the mandate for Workato AI Labs, and DigitalOcean delivered.

Workato's AI Lab hit a wall: no GPUs, no progress

Without dedicated infrastructure, their team couldn't advance critical AI research

Frame 2

Frame 3
Plus **a lower TCO**, eliminating redundant computation costs.

67% Lower Inference costs

Frame 4
"DigitalOcean was the fastest provider to get us up and running." — Oscar Wu, AI Research Scientist.

Frame 5
Purpose-built for Production AI.
Optimized NVIDIA HGX™ H200 clusters on DigitalOcean

Social Post Copy

Channel	Copy	Image preview	Comments
LinkedIn	When @Workato's AI Lab hit a wall without GPU infrastructure, major cloud providers offered weeks-long waits and complex procurement. We at DigitalOcean took a different approach: ✅ @NVIDIA H200 clusters deployed in days		

	✅ Multi-node optimization completed in hours by our solutions architects ✅ 2-3x faster time to value through expert support The impact: "DigitalOcean is the only provider that lets me focus on my job instead of fighting infrastructure." — Kevin Huang, AI Lab Engineer at Workato. Now their team is building self-improving AI agents for the autonomous enterprise—powered by optimized DigitalOcean infrastructure. Speed and expertise aren't nice-to-haves. They're the foundation of innovation.		
X	When @Workato's AI Lab needed GPUs, competitors had waitlists. #DigitalOcean delivered @NVIDIA H200 clusters in days. Our SAs optimized their multi-node setup in hours. The result: 2-3x faster time to value. Speed + expertise = breakthrough AI research		
Instagram			
Bluesky			
Threads			

Workato suggested copy

Suggested stakeholder amplification

NVIDIA suggested copy

Suggested stakeholder amplification

Technical social posts

Audience: Technical
DODX ticket
Social ticket

- **Inference cost:** $0.77 power 1M tokens – **67% lower**
- **Throughput:** 13,561 tokens per second per GPU – **67% higher**
- **Time-to-First-Token** (TTFT): 1,455 ms at high load – **77% faster**
- **Time-to-Value:** Reduced from weeks to days – **2–3X acceleration**

Deprecated -Technical Social Posts Draft

DEPRECATED

Technical Social Posts
STO: Jakub Lukaszczyk / Julia Downing
Contributors: Waverly Swinton Tayler Mehit Jakub Lukaszczyk Julia Downing TBD
Go Live Date / Time: March 11 at [TIME]

Audience: Technical
DODX ticket
Social ticket

March 11 Animation

Audience: Technical audience, AI natives

	Message	Copy
Frame 1	Technical	

Option A: Current version updated to the [v2] Technical Blog.

Intro: [Intro Logo "Workato Powered by DigitalOcean x NVIDIA]

Frame 1: *The Hook*
Workato's AI Labs had one question: does a smarter inference stack actually pay off?
Here's what they found

Frame 2: *Setup*
Model: Llama-3.3-70B (FP8)
Context: 100K tokens
Hardware: NVIDIA H200s
Platform: DigitalOcean Kubernetes (DOKS)
Test: NVIDIA Dynamo + vLLM vs. optimized vLLM baseline

Frame 3: *Results*
Same hardware. Same model
67% higher throughput per GPU
77% faster time-to-first-token
~~67% lower cost per 1M tokens~~
67% lower model cost using half the GPUs

Frame 4: *Why*
At 100K-token context, duplicate prefill is your biggest cost
Dynamo routes requests to the GPU that already has the context cached
Less recompute. Lower latency. Better economics

Frame 5: *CTA*
Read the Technical Blog for full benchmark + architecture breakdown

Option B: New version converging [v2] Technical Blog and "Social Posts Draft".

Intro: [Intro Logo "Workato Powered by DigitalOcean x NVIDIA]

Frame 1: *The Hook*
Workato processes 1 trillion automated workloads. Inference infrastructure has to keep up.

Frame 2: *Challenge*
Delivering agentic AI across thousands of enterprise customers means 100K-token context lengths and high concurrency. Adding GPUs alone doesn't solve it.

Frame 3: *Solution*
DigitalOcean's Agentic Inference Cloud routes every request to the GPU that already has the context cached.

- NVIDIA Dynamo

- vLLM
- H200 GPUs on DOKS

Frame 4: *Results*
The results on identical hardware:

- 67% higher throughput per GPU
- 79% lower end-to-end latency
- 77% faster time-to-first-token
- 67% lower model cost using half the GPUs

Frame 5: *CTA*
See how DigitalOcean helped Workato build inference infrastructure that scales with enterprise AI.

Social Post Copy

Channel	Copy	Image preview	Comments
LinkedIn	Workato's AI Labs wanted to know if cluster-level routing intelligence moves the needle at production scale. Spoiler alert: Routing requests to GPUs that already have the context cached cuts inference costs by 67%. They tested Llama-3.3-70B (FP8) across 1-, 2-, and 4-node DOKS clusters on NVIDIA H200s — comparing NVIDIA Dynamo + vLLM against a fully optimized vLLM baseline (prefix caching, chunked prefill, FP8 KV cache, all enabled). On the 2-node cluster, Dynamo delivered: ➡️ 13,561 tokens/sec/GPU vs. 8,111 — 67% higher throughput on identical hardware		

	➡️ 1,455ms median TTFT vs. 6,452ms under concurrency=32 — 77% faster ➡️ $0.77/1M tokens — 67% lower cost than a 4-node optimized vLLM setup The core insight: at 100K-token context lengths, KV cache reuse isn't a nice-to-have. Redundant prefill computation becomes the dominant cost driver. Dynamo eliminates it with state-aware, token-level routing and not round-robin.		
X	.@Workato's AI Labs ran a real benchmark. Not a toy workload—Llama-3.3-70B (FP8), 100K context, production-grade SLAs. Results on DOKS with NVIDIA Dynamo + vLLM vs. optimized vLLM baseline: ➡️ 67% higher token throughput per GPU ➡️ 77% lower TTFT at concurrency=32 ➡️ $0.77/1M tokens — 67% cheaper The delta isn't the model. It's KV cache-aware routing.		
Instagram			
Bluesky			
Threads			

Workato suggested copy

Suggested stakeholder amplification

NVIDIA suggested copy

Suggested stakeholder amplification

Phase 2

To be filled in - actions include:

- Co-event in Workato's SF office
- Joining their podcast/video series
- Onsite video featuring Oscar at Workato

ELT Social Guide

ELT Social Sharing Guide

[Reposting Guide](#)
- [On X](#)
- [On LinkedIn](#)

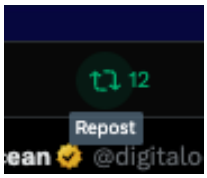
[Social Media Links](#)
[Story Talking points](#)

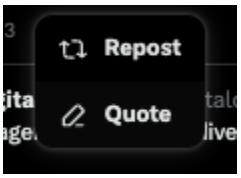
Reposting Guide

Amplify existing DigitalOcean posts by **sharing**, rather than publishing your own post, to improve brand engagement, centralize the conversation, and strengthen the brand association with executives.

On **X**

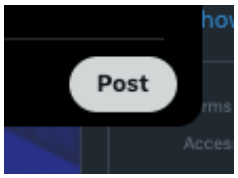
1. Click the **Repost** button located on the bottom left of the post and select **Quote**





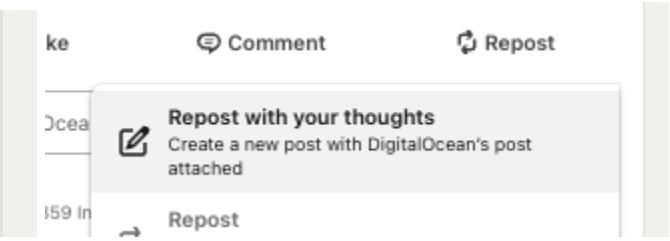
2. Leverage below *Talking Points* to reinforce DigitalOcean's messaging in your own voice

3. Click **Post** in the bottom right corner when you're ready to publish

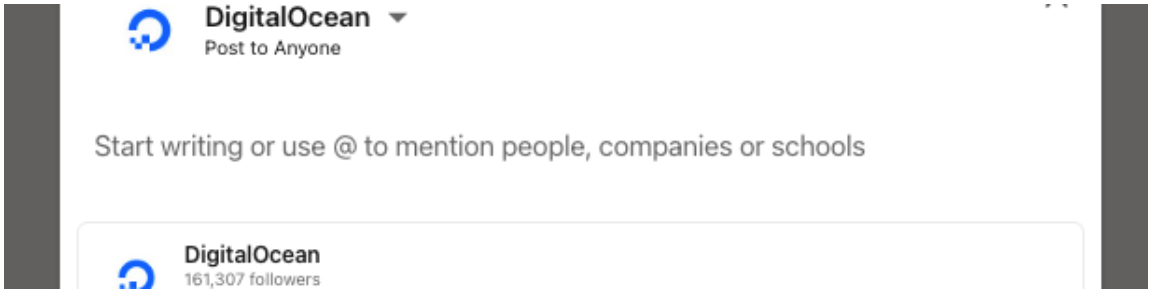


On **LinkedIn**

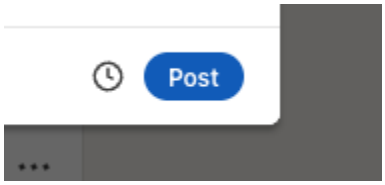
1. Click **Repost** button located on the bottom left of the post and select **Repost with your thoughts**



2. Leverage below *Talking Points* to reinforce DigitalOcean's messaging in your own voice
    i. Anything above 200 characters will get truncated to "see more…", ensure you get your point across early.



3. Click **Post** in the bottom right corner when you're ready to publish or select the clock to schedule it for later



Social Media Links

Channel	Link	Customer / Partner handles
LinkedIn	https://www.linkedin.com/feed/update/urn:li:activity:7434595794889977856	NVIDIA: Tag @nvidia in your X share once where it's mentioned Eg. "...configuration of @nvidia Dynamo" Workato: Tag @workato in your X share once where it's mentioned Eg., "Our partnership with @Workato…"
X	https://x.com/digitalocean/status/2028829760336388343	NVIDIA: Tag @nvidia in your LinkedIn share once where it's mentioned Eg. "...configuration of @nvidia Dynamo" Workato: Tag @workato in your LinkedIn share once where it's mentioned Eg., "Our partnership with @Workato…" gave

Story Talking points

Just the Stats

- **67% lower Inference cost:** $0.77 power 1M tokens; using half the GPUs
- **67% higher Throughput:** 13,561 tokens per second per GPU
- **77% faster Time-to-First-Token** (TTFT): 1,455 ms at high load
- **79% lower latency:** end-to-end latency
- **2X+ acceleration in Time-to-Value:** Reduced from weeks to days

Talking points

Summary

- Using DigitalOcean's Agentic Inference Cloud, Workato's AI Research Lab was able to achieve **significant improvements in performance at scale** (67% higher throughput, 77% faster time to first token, and 79% lower latency) as a result of the **inference-optimized architecture.**
- DigitalOcean's **simple operations** enabled the Workato AI Research Lab team to deploy in days vs weeks, **experiencing 2-3X acceleration in time to value.**
- They **reduced their total cost of ownership**, with 67% lower inference cost.

Details

- What does Workato AI Labs do?
    - The Workato AI Research Lab team is pioneering the future of enterprise automation by building self-improving, agentic AI systems that optimize large language models. This research directly enables the deployment of scalable AI solutions, driving core business functions in IT, HR, and sales.
- Why DO?
    - Workato's AI Research Lab uses DigitalOcean's Agentic Inference Cloud to advance the development of its next-gen enterprise AI Agents.
    - Once running on DigitalOcean, Workato achieved immediate gains on frontier models, including Llama-3.3-70B.
    - The results? 67% lower inference cost, with 67% higher throughput and 79% lower end-to-end latency. The Workato AI Lab reduced time to value from weeks to days
- How?
    - Inference-optimized architecture that works: NVIDIA Dynamo with KV-aware routing + vLLM on DigitalOcean Kubernetes
    - Workato's AI Lab achieved these efficiencies using NVIDIA Hopper GPUs, optimized on DigitalOcean's Inference Cloud. Learn more here [Link]

Optional Technical blog talking points

- Breakthrough Inference throughput -

    - Optimized configuration of NVIDIA Dynamo, on DigitalOcean's Inference Cloud achieved a 3.3x faster output (than other disaggregated configurations tested).
    - This demonstrates a superior data processing capability for LLMs for customers leveraging NVIDIA + DigitalOcean together.
- Superior Low-Latency Performance (TTFT)
    - Workato AI Research Lab's deployment dramatically improved the crucial time-to-first-token (TTFT) latency.
- Scalable Architecture for Enterprise AI
    - Workato AI Research Lab successfully benchmarked the Llama-3.3-70B model on DigitalOcean Kubernetes with NVIDIA H200 GPUs - enabling them to serve massive models at enterprise scale.